



SUSTAINABILITY IMPACTS OF AI

NIKOLA MILOJEVIC-DUPONT

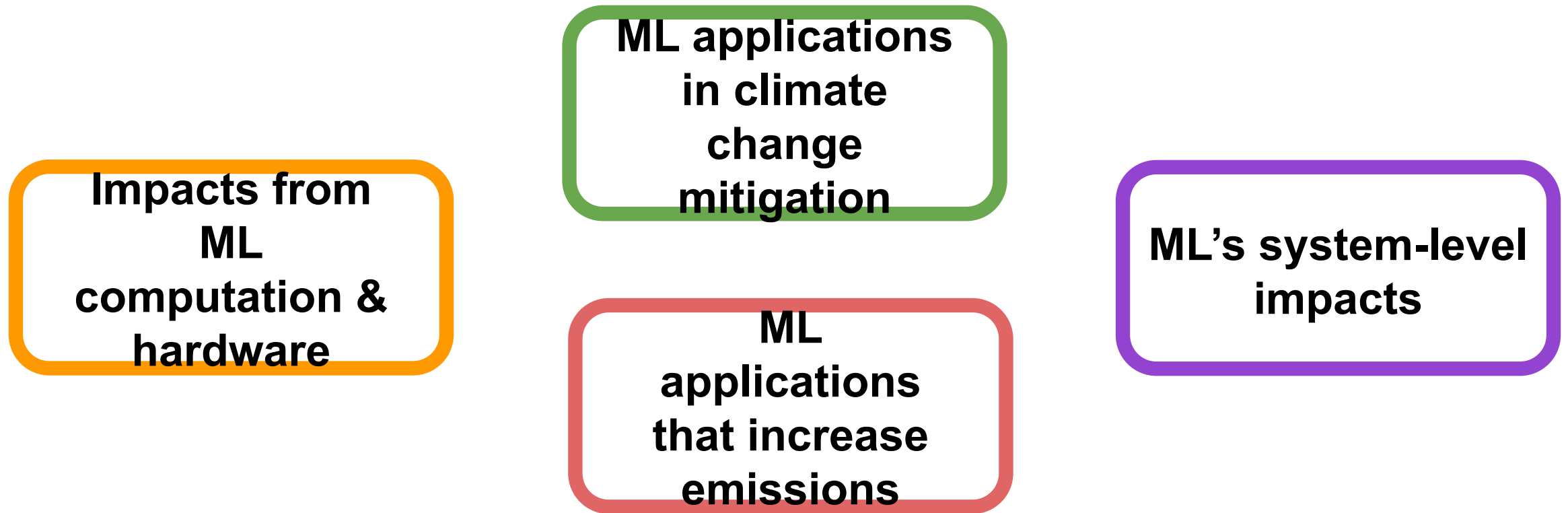
CLIMATE CHANGE AI SUMMER SCHOOL 2026

Image: Adrien Ob

ACKNOWLEDGMENTS

Thanks to **Priya Donti** and **Shaolei Ren** for inputs & feedback on this lecture, and **Lynn Kaack** for previous materials included!

CALLBACK: MULTIFACETED RELATIONSHIP BETWEEN ML AND CLIMATE



POLL ONE

Which one of four impact **categories** do you believe may have the **most** substantial **impact** on climate currently?

Impacts from
ML
computation &
hardware

ML applications
in climate
change
mitigation

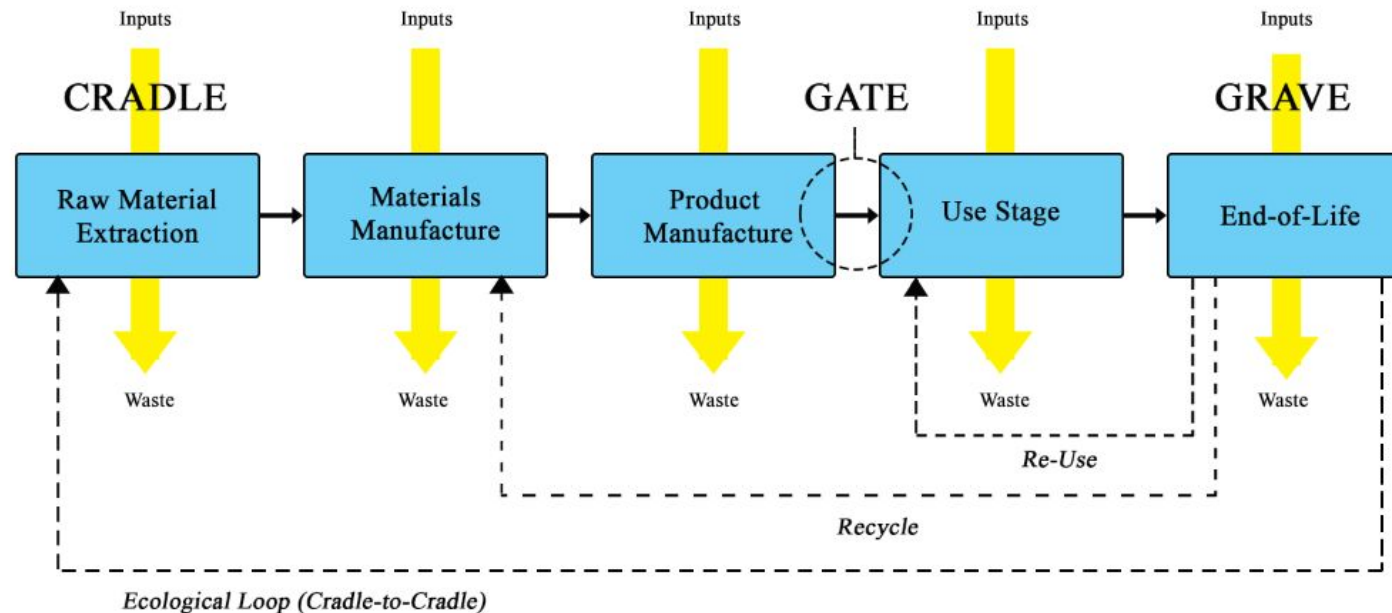
ML
applications
that increase
emissions

ML's system-level
impacts

LIFE-CYCLE ASSESSMENT

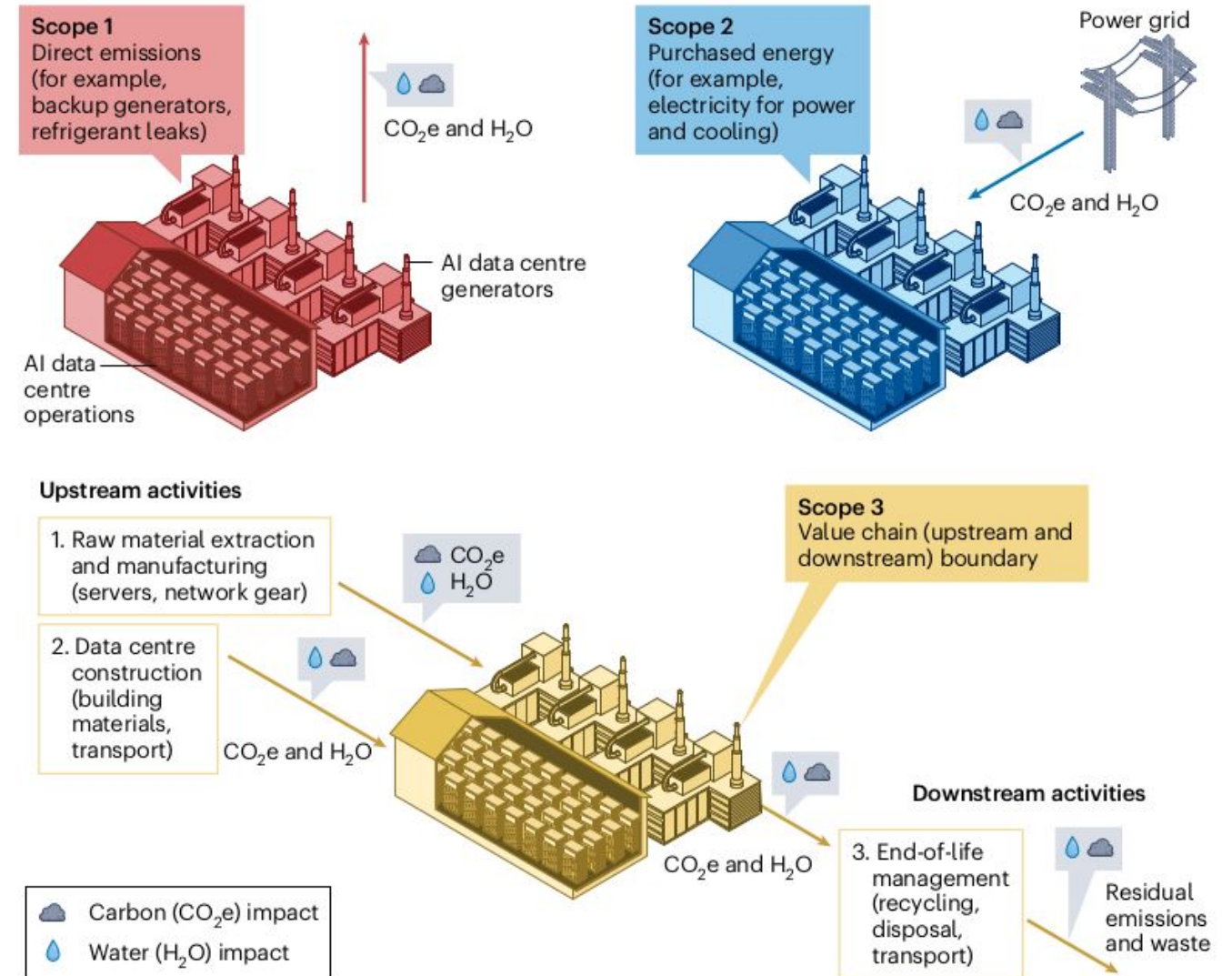
Whole value chain from **cradle** (raw material extraction), through use phase, to **grave** (disposal)

Assessing the **many** environmental impacts

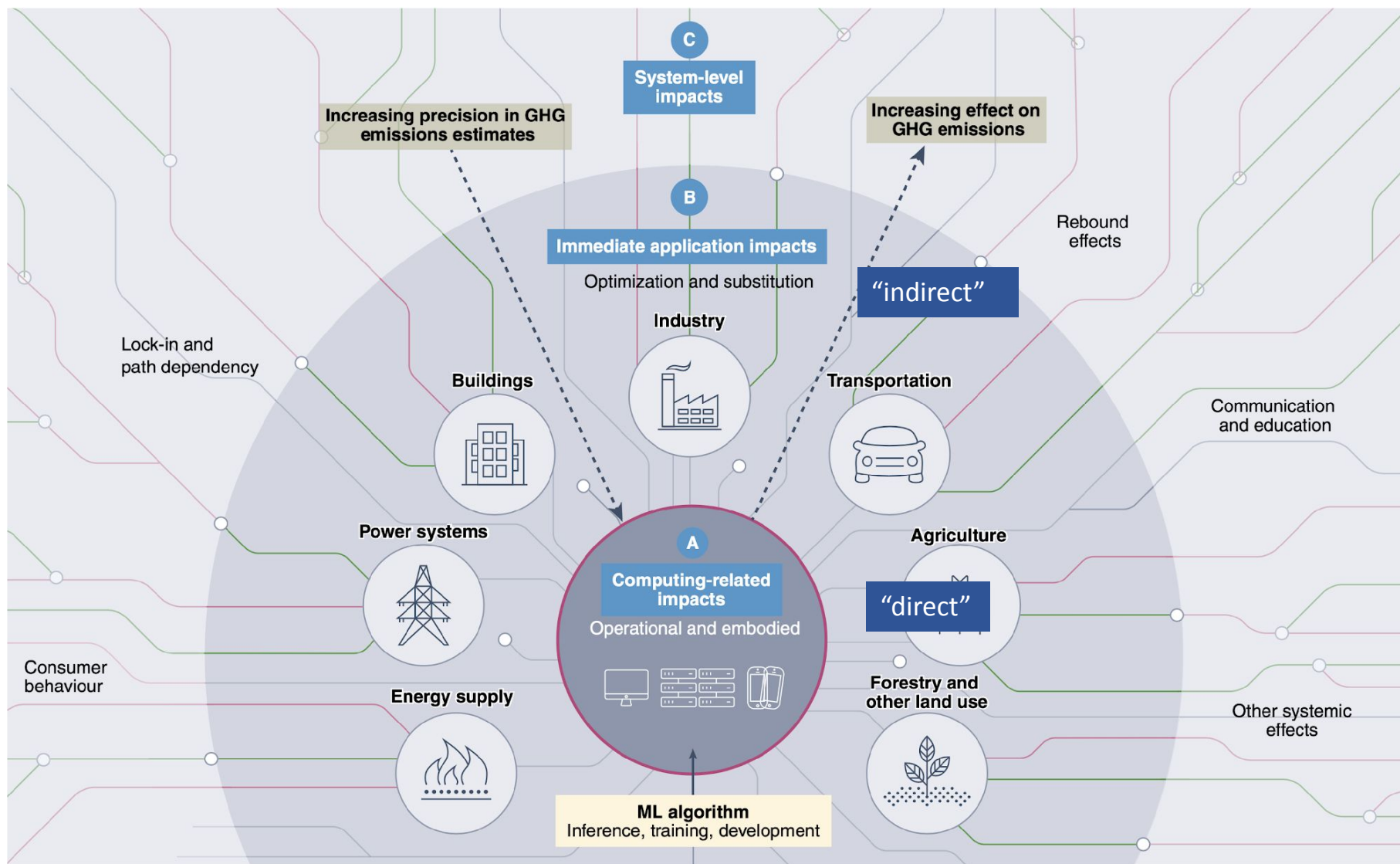


INVENTORY ANALYSIS: GHG PROTOCOL

- Global standardized frameworks to measure GHG emissions from private and public sector operations, value chains and mitigation actions
- **Scope 1:** Emissions from direct use of fossil fuels, e.g. for heat, fleet vehicles
- **Scope 2:** Emissions from purchased electricity and heat
- **Scope 3:** Emissions from value chains including customers
- Also extended to water



How AI Affects Greenhouse Gas Emissions



LECTURE OUTLINE

Computing-related impacts

Application impacts

System level impacts

Overarching considerations

Regulating energy and resource usage of AI

LECTURE OUTLINE

Computing-related impacts

Application impacts

System level impacts

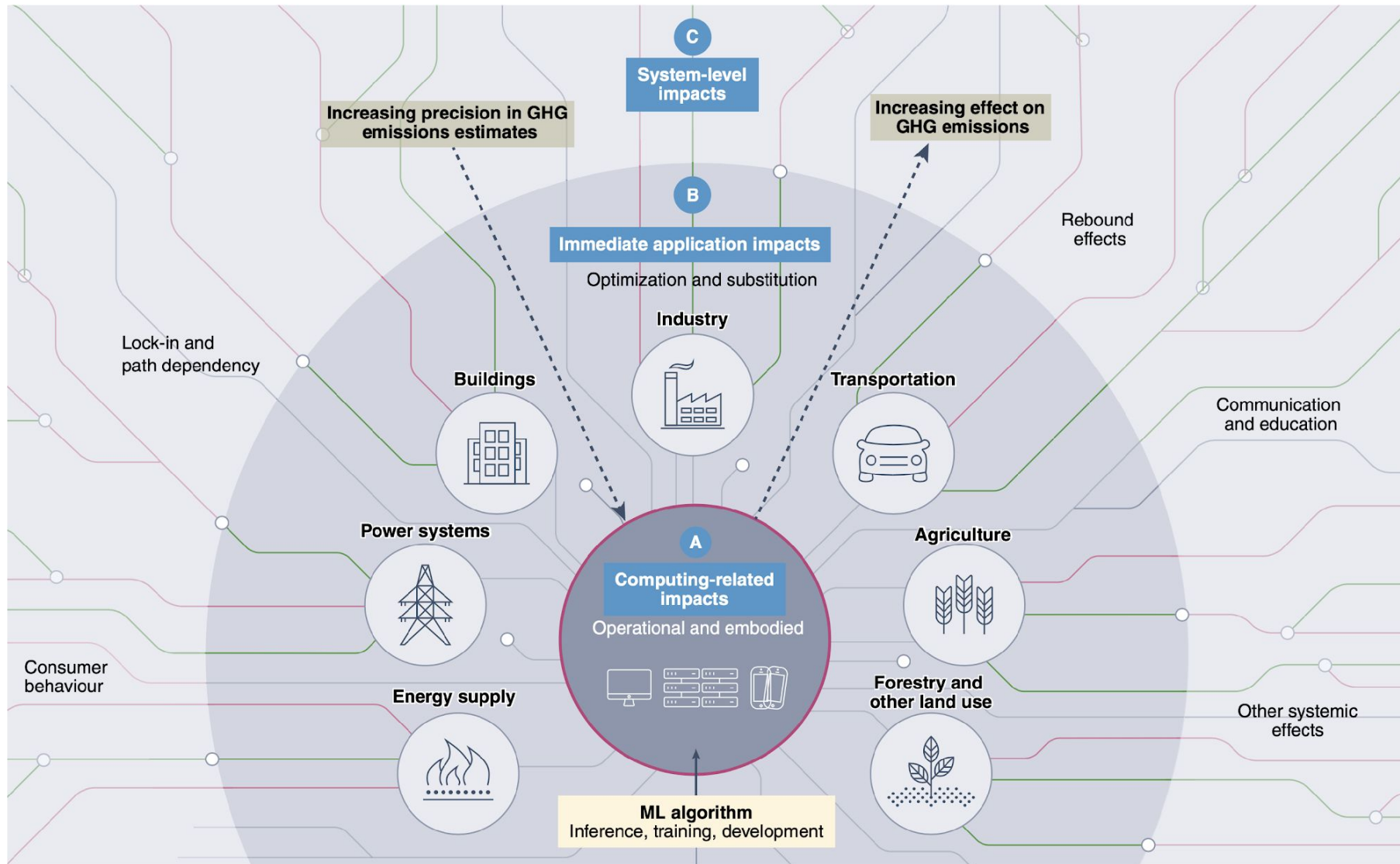
Overarching considerations

Regulating energy and resource usage of AI

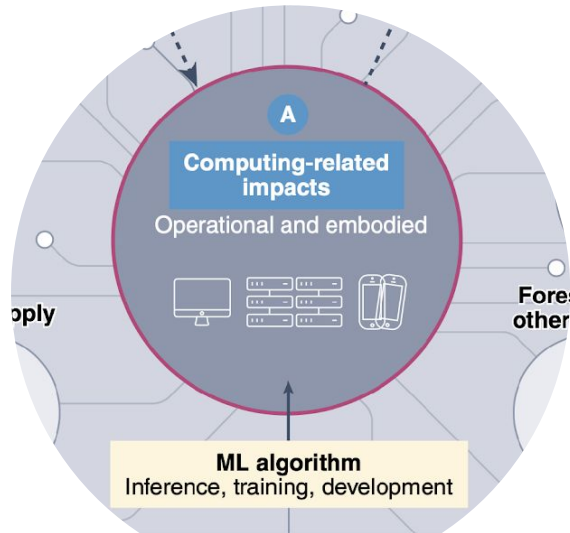
Training,
Inference

Impacts
differentiated
by tasks and
models

How AI Affects Greenhouse Gas Emissions

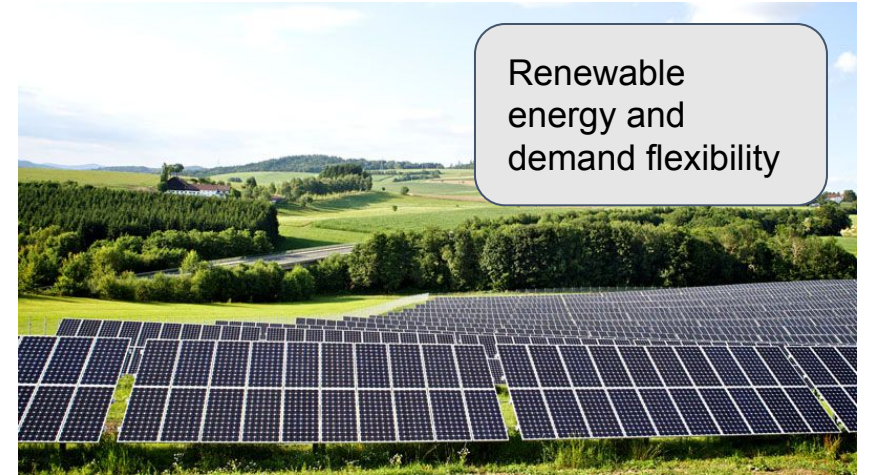


How AI Affects Greenhouse Gas Emissions

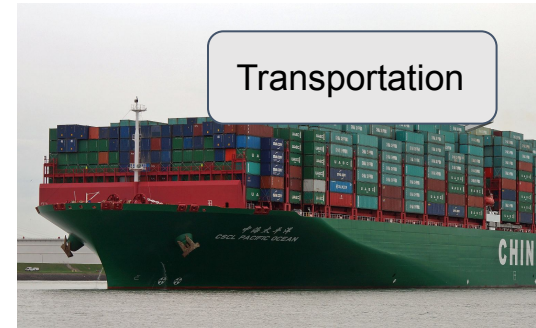


IMPACTS FROM ML COMPUTATION & HARDWARE

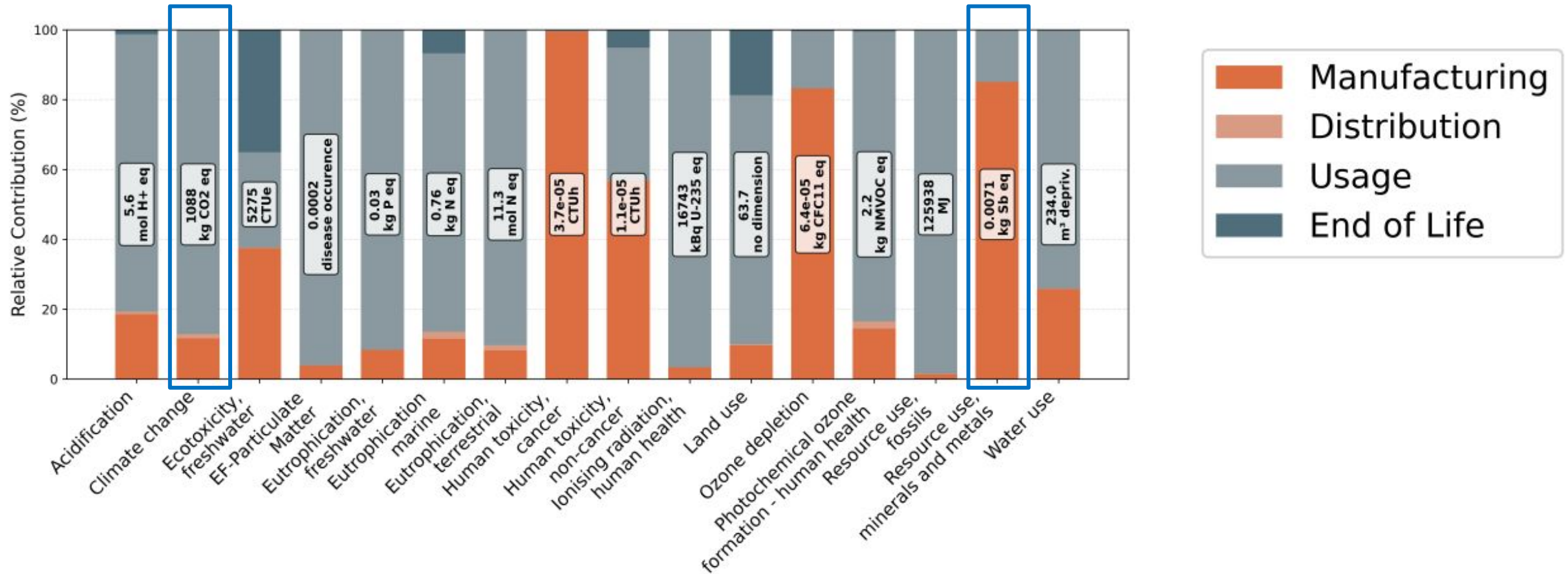
Operational emissions from energy consumed during computation



Embodied emissions from production and end-of-life of hardware

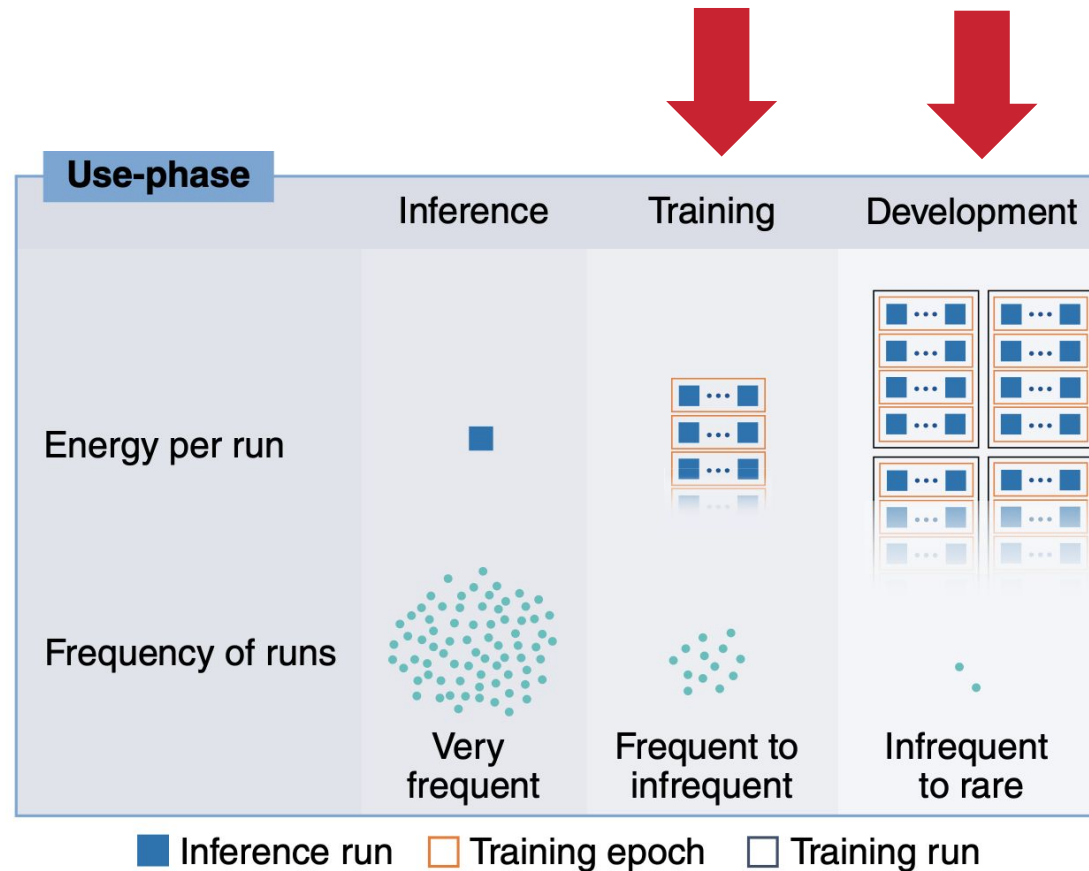


SUSTAINABILITY IMPACTS OF A GPU CARD



- Operational emissions >> Embodied emissions
- Importance of mining

COMPUTING -RELATED EMISSIONS FROM ML: DEVELOPMENT & TRAINING



REMINDER:

AI is not just “one thing”

Models have different sizes



MODEL SIZE INFLUENCES THE ENERGY DEMAND FOR TRAINING

~ 2 days of powering Berlin

Parameters and electricity consumption evaluations for several leading LLMs.

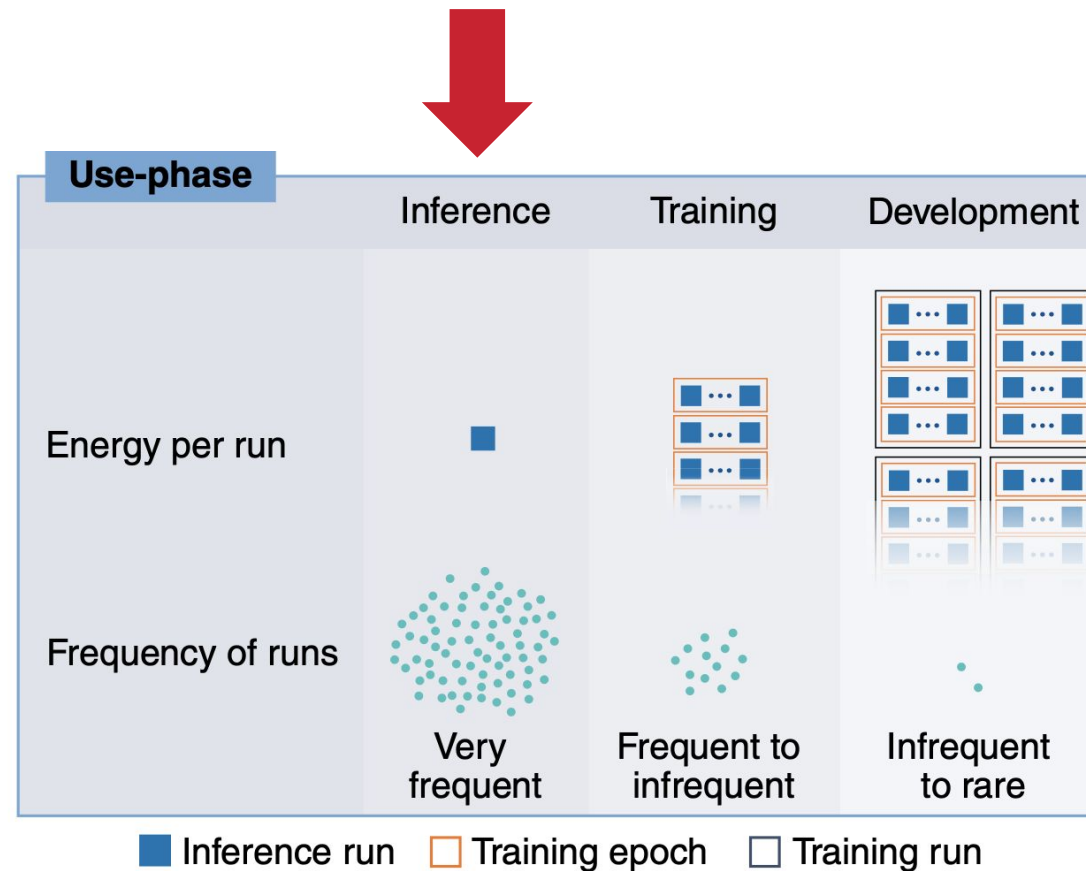
Developer	Model	Released year	Number of parameters (billion)	Accelerator type	Electricity consumption (MWh)	CO ₂ emissions (ton)	Literature
Google	BERT	2018	0.1	GPU V100	1.5	0.7	[13,14]
	GLaM	2021	1162	TPU v4	456	40	[19]
	PaLM	2022	540	TPU v4	3436	271.4	[38]
OpenAI	GPT-2	2019	1.5	TPU v3	1.7	0.7	[14,39]
	GPT-3	2020	175	GPU V100	1287	552	[32,40]
	GPT-4	2023	1800	GPU A100	51,772 to 62,318	12,456 to 14,994	[41,42]
Meta	OPT	2022	175	GPU A100	324	75	[43]
			7		74	31.2	[44]
			13		147	62.4	[44]
			70		688	291.4	[44]
	LLaMA 3	2024	1	GPU H100	/	107	[45,46]
			3		/	133	[45,46]
			8		/	390 to 420	[45,46]
BigScience	BLOOM	2022	70	GPU A100	/	1900 to 2040	[45,46]
			405		/	8930	[45,46]
			1		159	9.1	[18]
			104		267	15.2	[18]
DeepMind	Gopher	2021	176	TPU v3	433	24.7	[18]
			280		1151	380	[13,47]
					/	352	[4]
TII	Falcon	2023	40	GPU A100	163 ^b	8 to 90 ^b	[48–50]
			180		2,940 ^b	147 to 1,617 ^b	[48–50]
Mistral AI	Mixtral 8 × 7B	2024	47 ^a	GPU A100	232 ^b	6.3 ^b	[51–53]
DeepSeek	DeepSeek-V3	2024	671 ^a	GPU H800	1,025 ^b	550 ^b	[54,55]

^a These values denote the total number of parameters, with different quantities activated depending on the model architecture.

^b The electricity consumption data of some LLMs lack direct values from open-published literature. Thus, these data are estimated, and the estimation process is provided in [Appendix Table A1](#).

COMPUTING-RELATED EMISSIONS FROM ML:

INFERENCE



INFERENCE EMISSIONS ARE GROWING

- Both Meta and Google report that inference emissions outweigh those of training
- Depending on use, training or inference may dominate
- Consumers need transparency to trade off between models



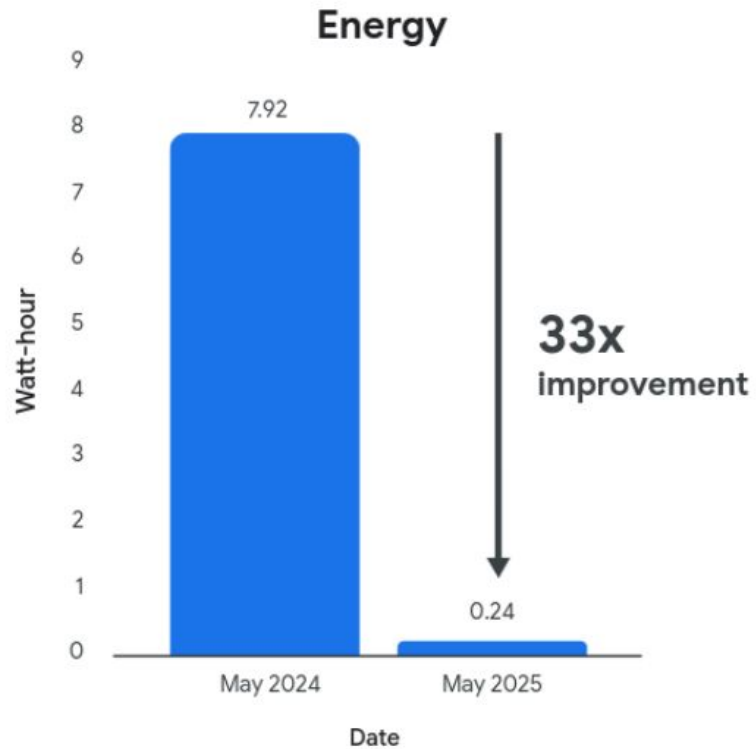
Model		Request freq.	Energy (kWh)	# inf. to equal training
Llama 3.1	8B	∞	0.011	276B
		8	0.051	59.5B
		1	0.333	9.12B
OLMo 2	7B	∞	0.018	20.9B
		8	0.049	7.68B
		1	0.348	1.05B
OLMo 2	13B	∞	0.033	22.1B
		8	0.057	12.8B
		1	0.386	1.89B



HIGHER EFFICIENCY ... LARGELY OUTWEIGHED BY USAGE GROWTH

Gemini efficiency improvements

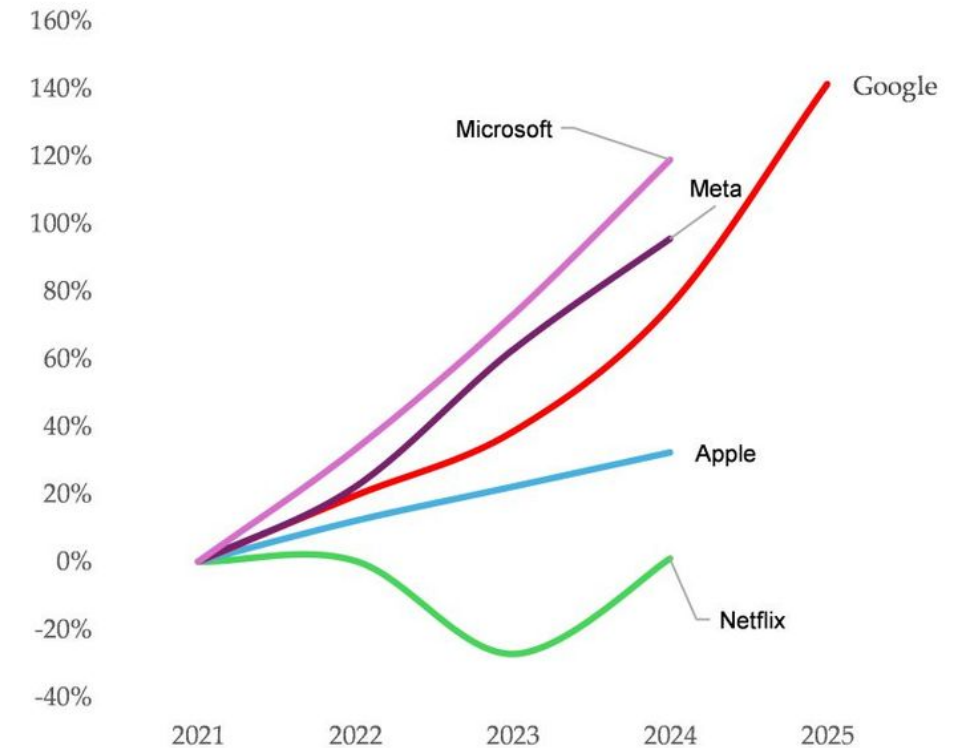
Energy and carbon efficiency improvements per median Gemini Apps text prompt from May 2024 to May 2025.



Rebound effect



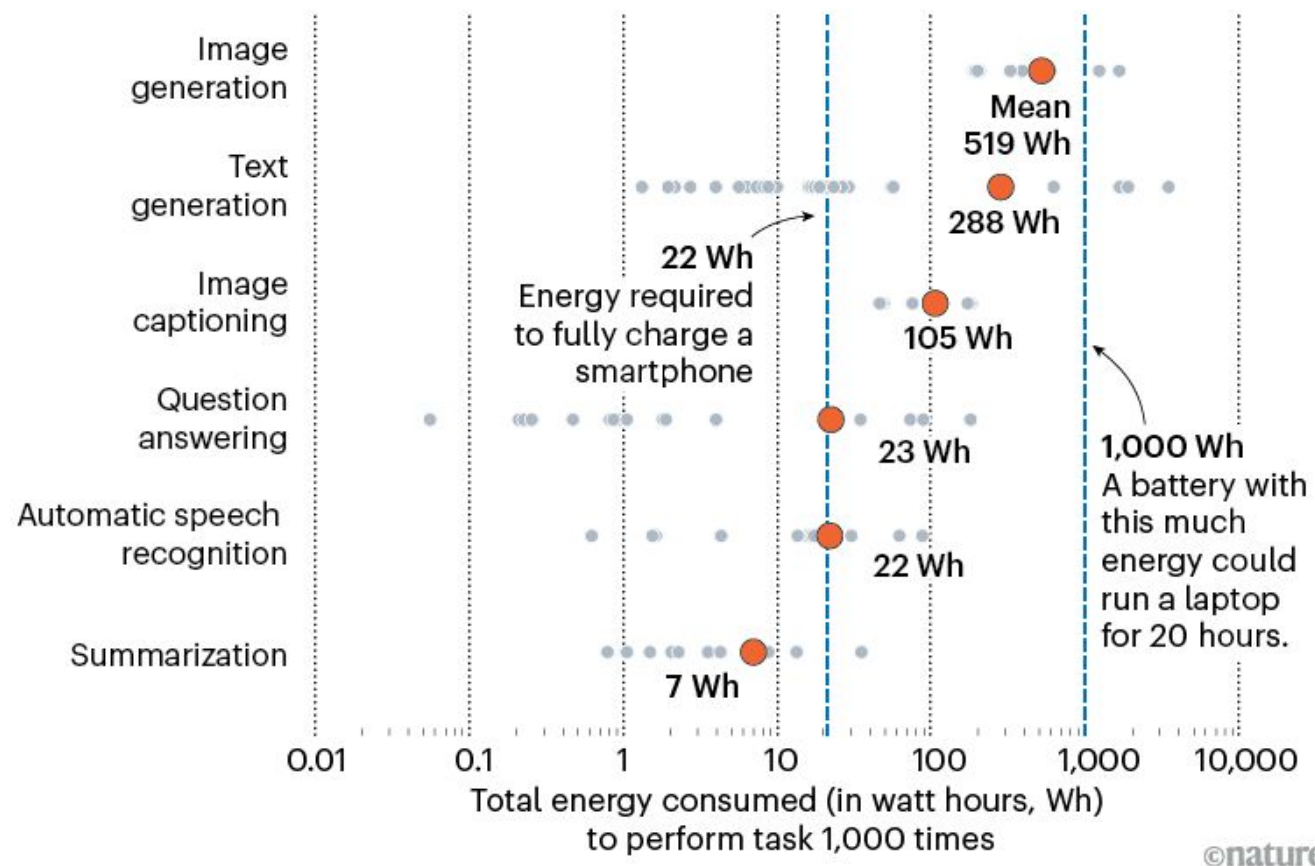
Percentage change in energy consumption from 2021



LARGE DIFFERENCES IN INFERENCE ENERGY NEEDS ACROSS TASKS

HOW MUCH ENERGY DOES AI USE?

The AI Energy Score project tested dozens of artificial-intelligence models to estimate how much energy they consume when performing various tasks. Plotting the energy required to perform a task 1,000 times shows that energy use varies greatly depending on the task and the model.



POLL Two

What is the order of magnitude between
one **prompt**
and one **8-sec medium-definition video**?

a) x **10**, b) x **100**, c) x **1,000**, d) x **10,000**



Lights, Camera, Carbon: Architectural Scaling Laws for Video Generation Energy Consumption

Nidhal Jegham, Boris Gamazaychikov, Sasha Luccioni

Sora 2.0 Pro is gone.
Here's what it cost to run.

OpenAI discontinued Sora in March 2026. We estimated its energy footprint from architecture and API timing alone, with no access to model weights required.

Per 8-second, 720p video generated

418.5 Wh

≈ 1,700× the energy of one Gemini text prompt (0.24Wh)

602.6 GWh

estimated if 4M users generated 2
videos/day for 6 months

~113,000

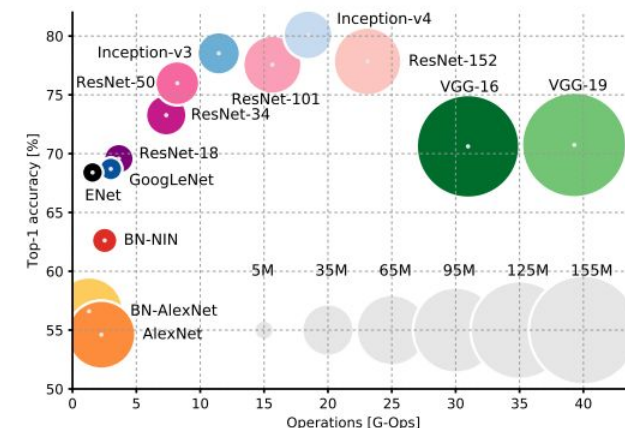
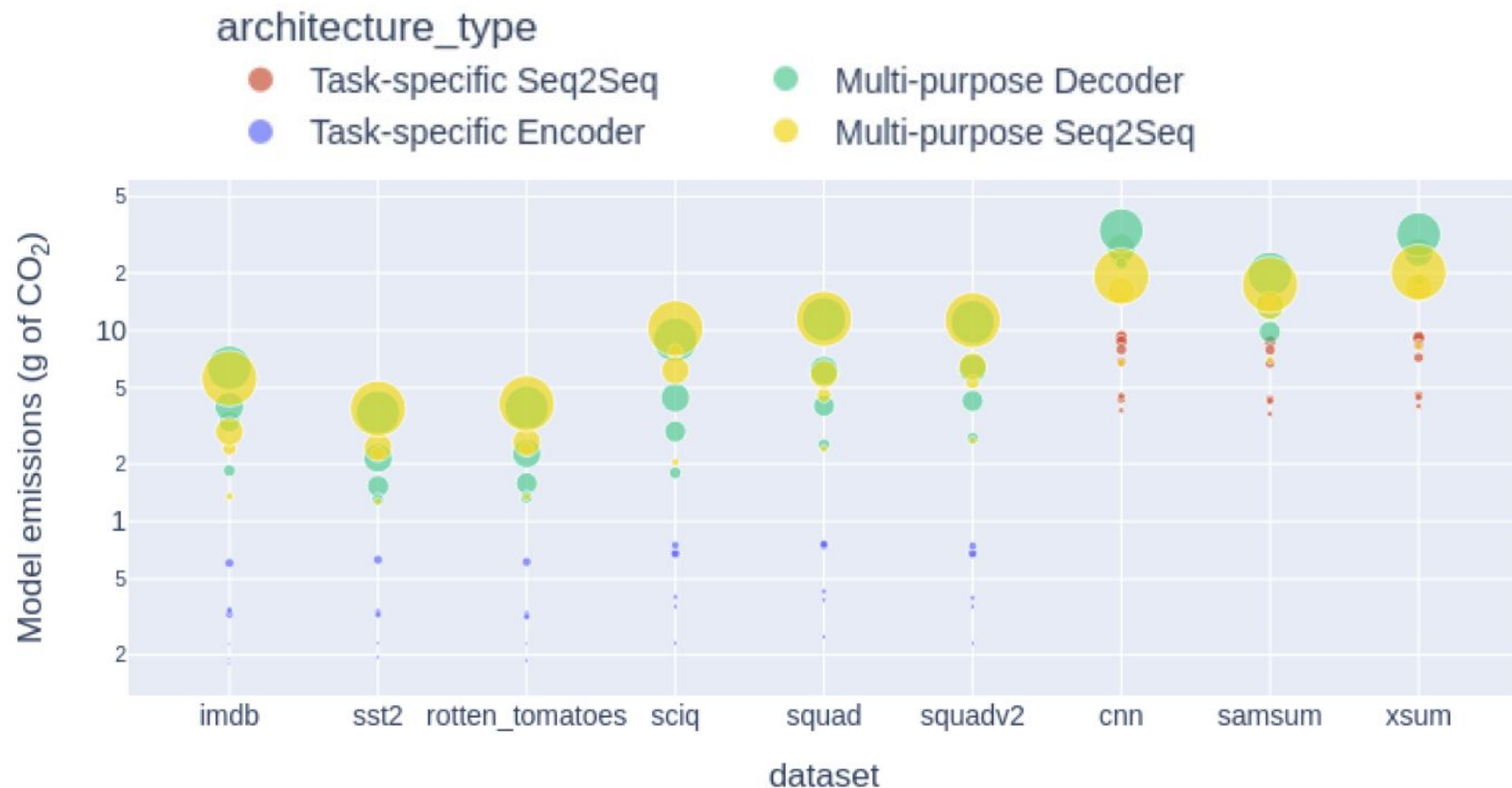
US households' electricity use
during the same period, equivalent

AI WITH LITTLE ENERGY USE Is COMMON & POSSIBLE

→ FRUGAL AI

Common Myth	Strategic Reality	Takeaways
Frugal AI = Cheap, Low Quality	Frugal AI prioritizes efficient, high-impact solutions, not cutting corners.	Reframe cost-saving as value-maximization through smarter, leaner AI investments.
Low-tech can't scale	Simple tools (WhatsApp, SMS, low-code AI) power platforms used by millions globally.	Embrace context-appropriate scaling – not every solution needs enterprise-grade infrastructure.
Only big tech can do AI	AI is increasingly democratized; startups and small teams build effective, lean models.	Tap into ecosystem partnerships and pilot frugal solutions for niche or underserved markets.
Frugal = Outdated Tech	Frugal AI leverages modern techniques (edge AI, model compression, efficient models).	Integrate Frugal AI metrics (TCO, energy score) into procurement and solution architecture.

USING GENERAL-PURPOSE MODELS FOR INFERENCE IS OFTEN INEFFICIENT COMPARED TO TASK-SPECIFIC MODELS



More parameters
→ more energy

Luccioni, Alexandra Sasha, Jernite, Yacine, and Strubell, Emma. "Power Hungry Processing: ⚡ Watts ⚡ Driving the Cost of AI Deployment?" *arXiv preprint arXiv:2311.16863* (2023)

MORE DATA $\neq$ ALWAYS BETTER PERFORMANCE

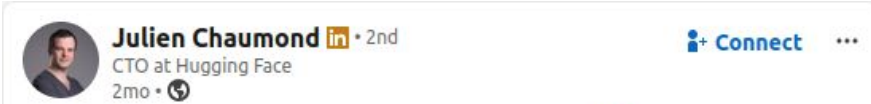
Table 6.1

Overall test set performance of local and global models (the best prediction performance on each local test set is highlighted in bold font).

Model	Trained on	Training epochs	Training samples	Training effort	Test RMSE ($\downarrow$) on		
					Stanford	SIRTA	DEWA
SUNSET	Stanford	21.2 $\pm$ 6.1	125,876	2.67 $\pm$ 0.77M	2.68	136.96	155.82
	SIRTA	9.6 $\pm$ 1.8	438,172	4.21 $\pm$ 0.81M	6.53	96.42	219.79
	DEWA	11.8 $\pm$ 2.2	85,278	1.01 $\pm$ 0.19M	4.65	160.28	89.49
	Stanford+SIRTA	28.6 $\pm$ 5.6	564,048	16.13 $\pm$ 3.15M	2.61	96.34	129.21
ConvLSTM	Stanford	11.7 $\pm$ 2.7	125,876	1.47 $\pm$ 0.34M	2.62	113.05	N/A
	SIRTA	5.6 $\pm$ 2.6	438,172	2.49 $\pm$ 1.13M	4.39	96.46	N/A
	Stanford+SIRTA	6.0 $\pm$ 1.7	564,048	3.36 $\pm$ 0.99M	2.62	95.53	N/A

Paper also shows that **fine-tuning a global model on DEWA data** (all or big enough subset) yields **similar performance** for DEWA compared to the **local DEWA model** (but with higher total training effort)

TOWARDS LOCAL MODELS



This is where we are right now. And it feels pretty magical 🦄

Qwen3.6 27B running inside of Pi coding agent via Llama.cpp on the MacBook Pro

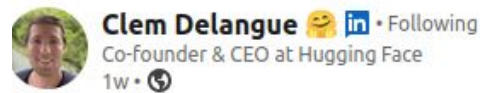
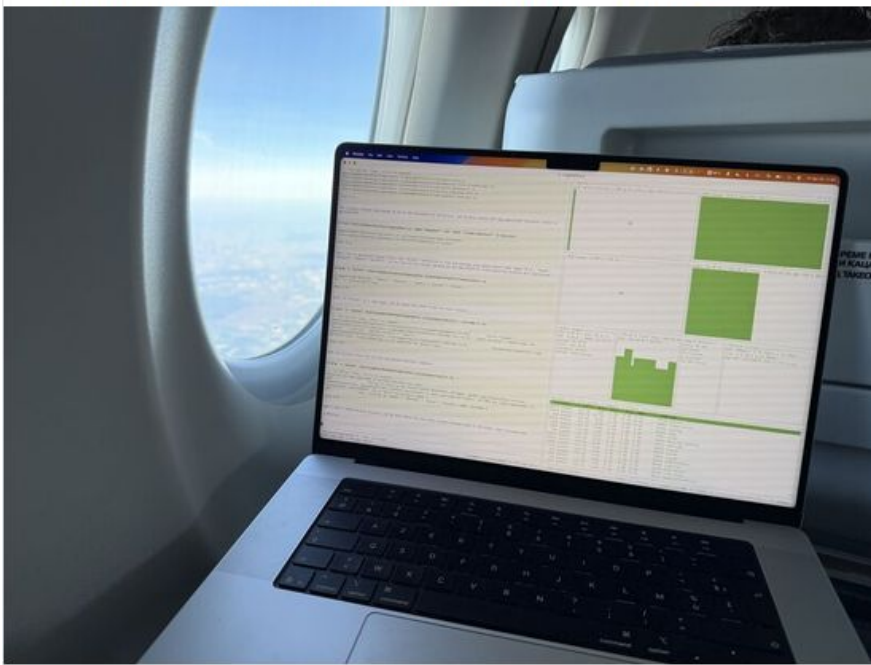
For non-trivial tasks on the [Hugging Face](#) codebases, this feels very, very close to hitting the latest Opus in Claude Code, or whatever shiny monopolistic closed source API of the day is.

In full airplane mode.

Most people haven't realized this yet.

If you have, it means you have a huge headstart to what I call the second revolution of AI.

Powerful local models for efficiency, security, privacy, sovereignty 🔥



A study from [Stanford University](#) showed that 71.3% of chatgpt queries could be accurately answered by a local model. I suspect a major part of enterprise AI workloads could be run locally too for free (compared to the massive costs of frontier API cost).

Also, it reduces the risk of these workloads being taken away from you because you own the models instead of renting them - which sounds like a good idea these days haha.

That's why we're introducing the ability for everyone to filter AI models on [Hugging Face](#) based on your local hardware.

For me, there are 800k+ public models that fit on my M5 24GB and that I can use easily thanks to llamacpp.

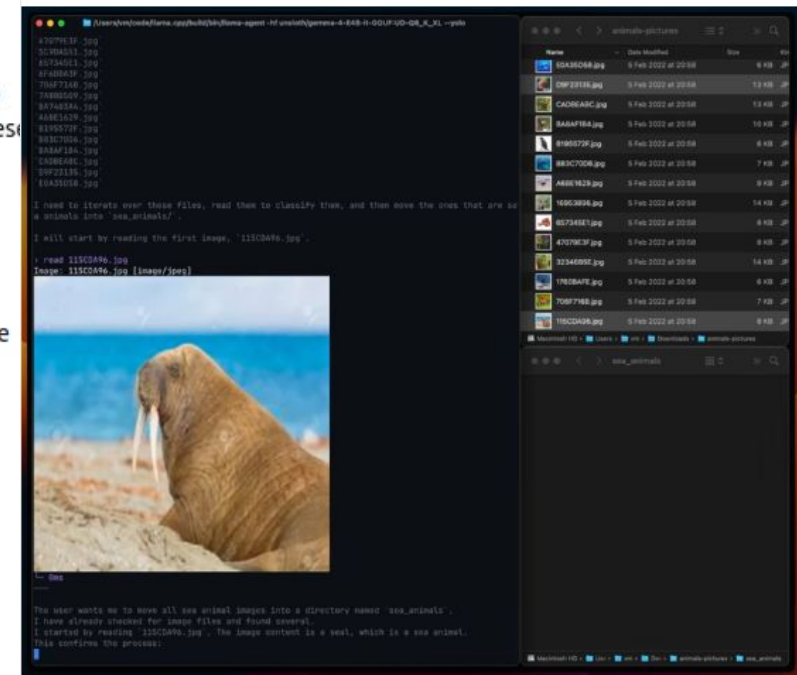
Let's go local AI!



I just watched the latest Google Gemma 4 open model work on my laptop. No cloud. No API key. No catch.

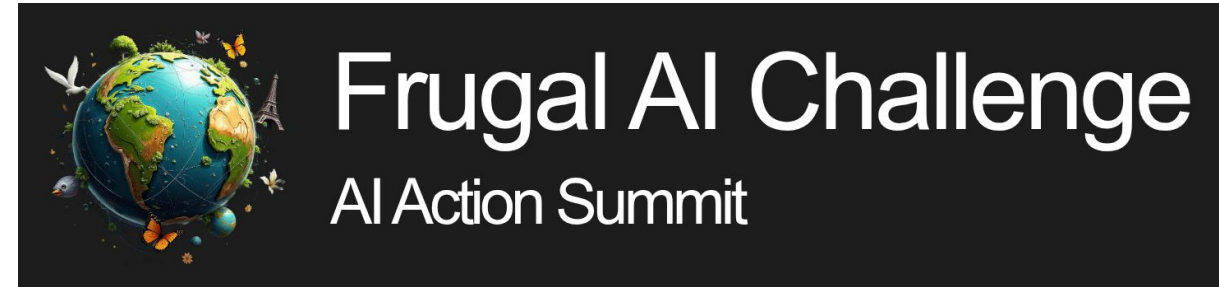
I pointed Gemma 4 E4B at a folder of mixed photos and asked it to sort the sea animals into their own directory. It opened each image, identified what it saw, and moved the files. All locally, in one agentic session, on a model small enough to fit in 5GB of RAM.

Three years ago this needed a data center. Today it runs while you're offline on a plane.

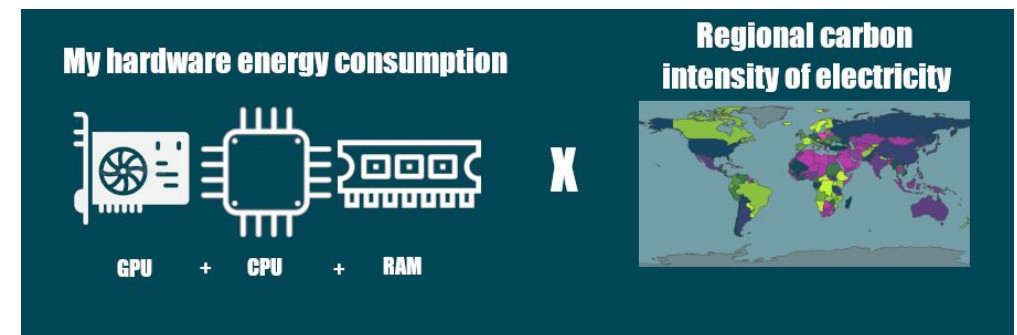
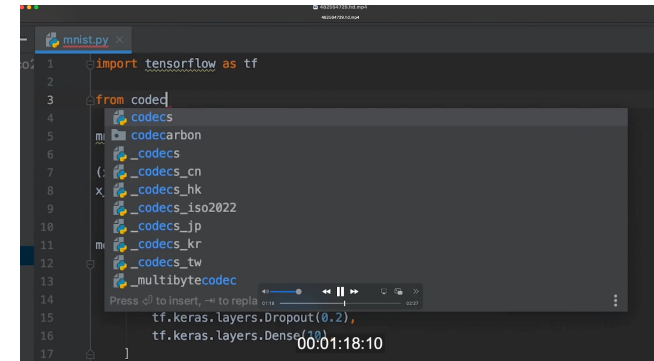


FRUGALITY IN AI DEVELOPMENT

- Frugal AI Challenge: Encourage academic and industry actors to keep efficiency in mind when deploying AI models
- Tracking energy consumption **and** performance
- Incentivize frugality in AI deployment while also addressing real-world challenges.
- Tasks:
 - Detecting climate disinformation
 - Classifying regions at risk of wildfires,
 - Detecting illegal deforestation.



Code Carbon



LAYERS AROUND THE MODEL ALSO MATTER

→ ESCAPING “TOKEN MAXXING” VIA SKILLS, ETC.



Ponytail

He says nothing. He writes one line. It works.

stars 82k release v4.8.4 npm v4.8.4 works with 20 agents license MIT

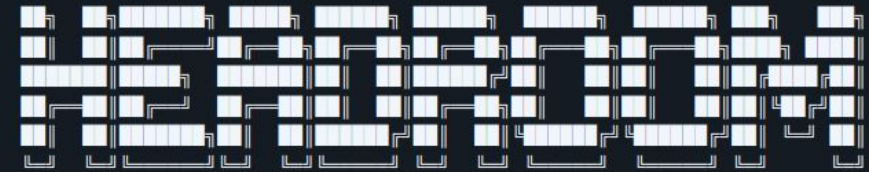
 TRENDSHIFT
#1 Repository Of The Day

 TRENDSHIFT
#1 Repository Of The Week

~54% less code (up to 94%) · ~20% cheaper · ~27% faster · 100% safe

Measured on real Claude Code sessions editing a real open-source repo (FastAPI + React), against the same agent with no skill. ~54% is the mean across 12 feature tasks (Haiku 4.5, n=4); it reaches 94% where an agent over-builds (a date picker) and is near zero where the code is already minimal. ponytail keeps every safety guard while a bare “write one-liners” prompt drops one. (The earlier single-shot benchmark reported 80-94% as a flat figure; against a fair agentic baseline that is the per-task ceiling, not the average.) [Full writeup](#) · [reproduce it](#).

[Español](#) · [한국어](#)



The context compression layer for AI agents

60–95% fewer tokens (for JSON data), 15-20% fewer tokens (for coding agents) · library · proxy · MCP · content-aware compressors · local-first · reversible

CI failing codecov 65% pypi v0.31.0 npm v0.22.4 model Kompress-v2-base license Apache 2.0 docs online

[Docs](#) · [Install](#) · [Proof](#) · [Agents](#) · [Discord](#) · [llms.txt](#)

AI agents / LLMs: read [/llms.txt](#) here, or fetch [the live index](#) / [full docs blob](#).

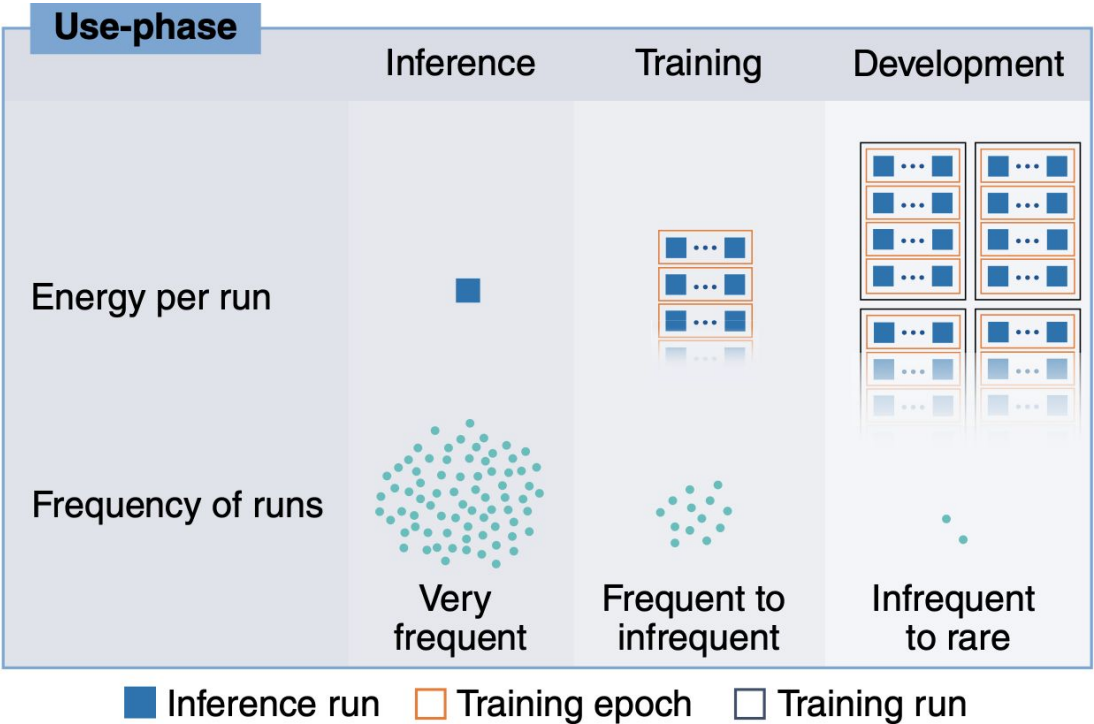
 1 GITHUB TRENDING
#1 Repository Of The Day

TINYML: RUNNING AI ON MICROCONTROLLERS

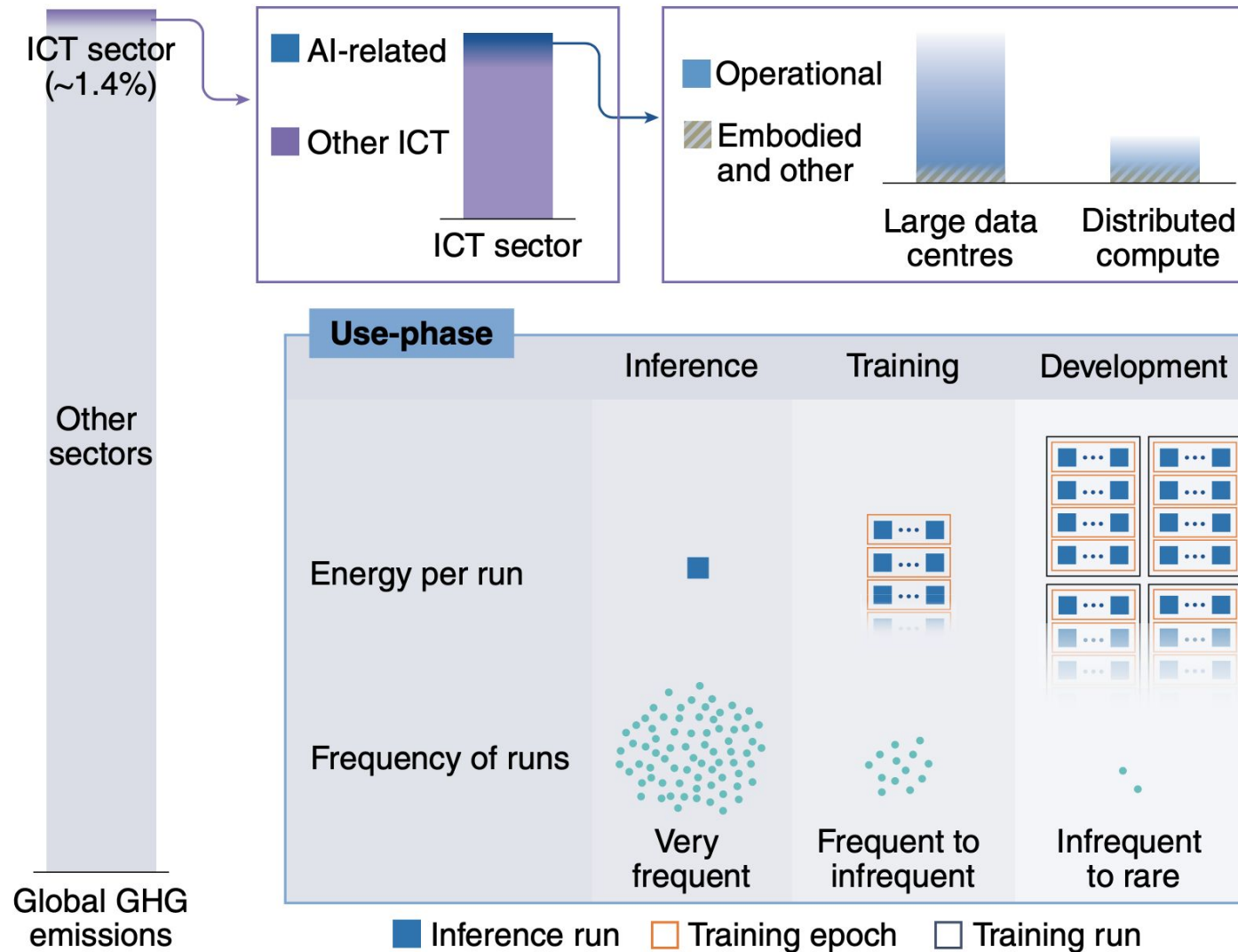
	Cloud AI (NVIDIA V100)	➔	Mobile AI (iPhone 11)	➔	Tiny AI (STM32F746)		ResNet-50	MobileNetV2	MobileNetV2 (int8)
Memory	16 GB	$\xrightarrow{4\times}$	4 GB	$\xrightarrow{3100\times}$	320 kB	← gap →	7.2 MB	6.8 MB	1.7 MB
Storage	TB~PB	$\xrightarrow{1000\times}$	>64 GB	$\xrightarrow{64000\times}$	1 MB	← gap →	102MB	13.6 MB	3.4 MB



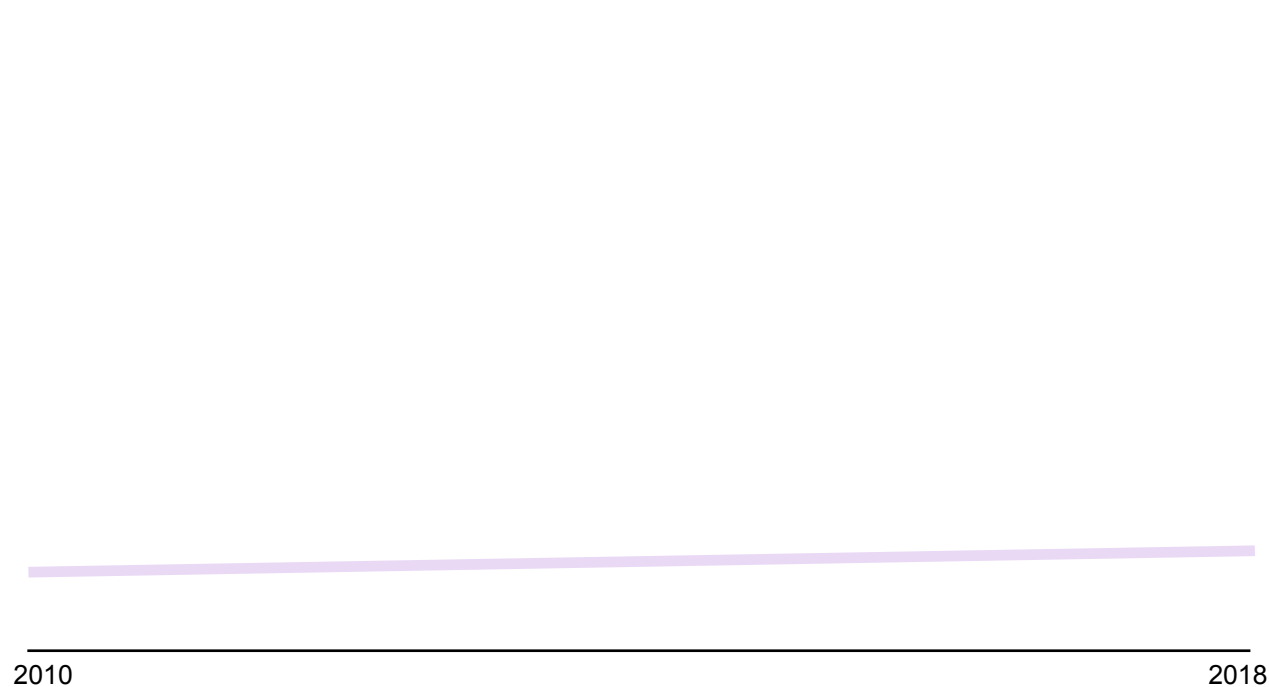
MOVING FROM BOTTOM-UP...



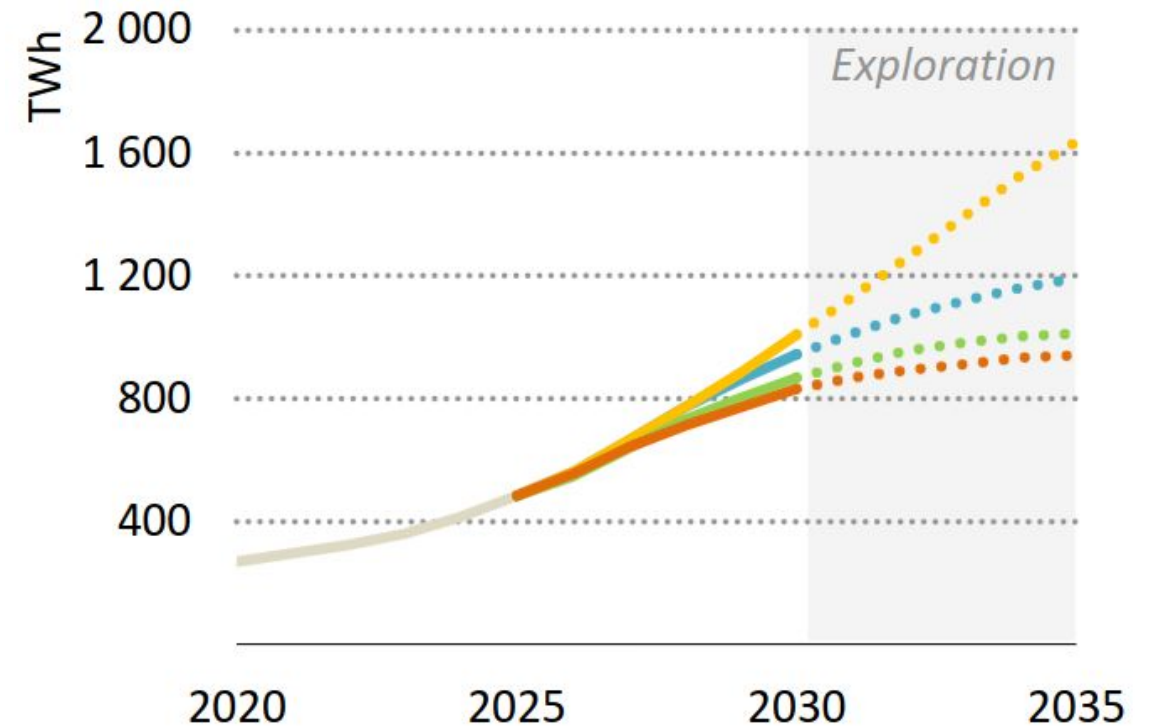
...TO TOP-DOWN PERSPECTIVE



ELECTRICITY DEMAND IS RAPIDLY GROWING



6% increase
between 2010 and 2018 (despite 5.5x increase
in compute instances)



250-500% increase
between 2020 and 2035

POLL THREE

Which **energy source** is expected to be the **most rolled out** to **meet the increased demand** in electricity from AI?

a) **renewables**, b) **coal**, c) **gas**, d) **nuclear**

ADDITIONAL ELECTRICITY SUPPLY:

CARBON INTENSITY?

COMPETING USES?

ASSOCIATED IMPACTS?



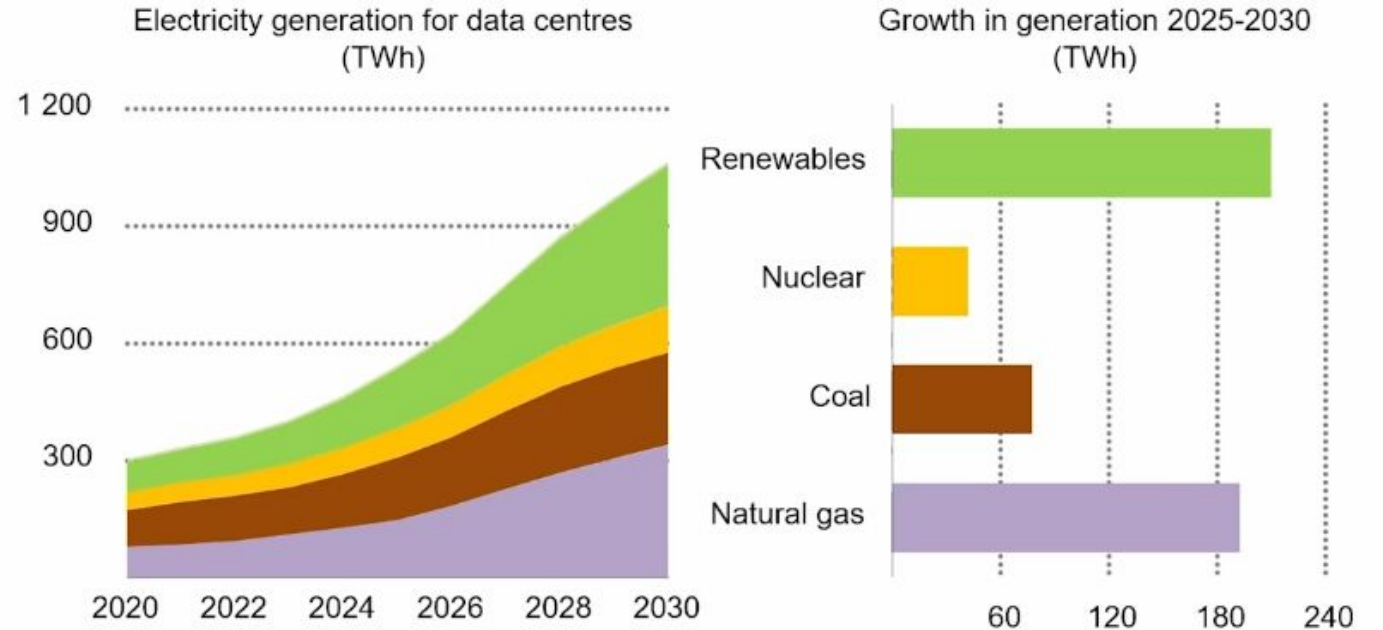
COMPANIES / ENVIRONMENT

Frontier Raises Over \$900 Million for Carbon Removal Investments from Google, Anthropic and Others

ENERGY

Chevron to fuel massive Microsoft data center in Texas using natural gas

PUBLISHED MON, JUN 22 2026 7:59 AM EDT | UPDATED MON, JUN 22 2026 3:19 PM EDT



GRID INTEGRATION OF DATA CENTERS: SPIKY POWER CONSUMPTION

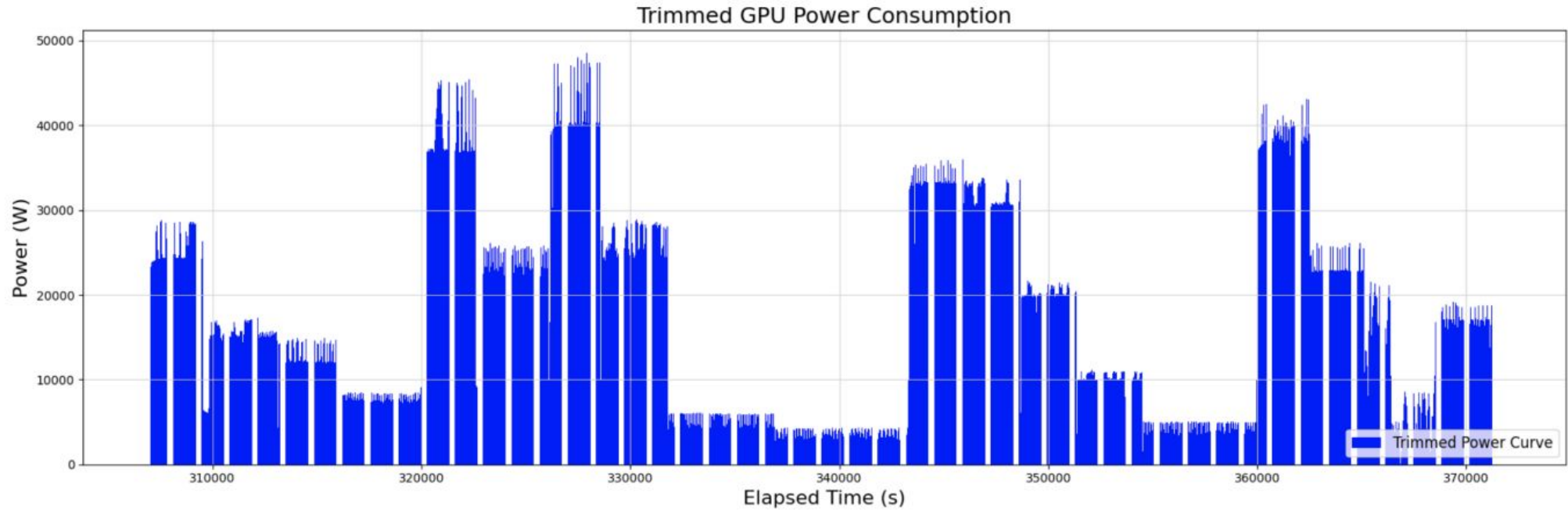


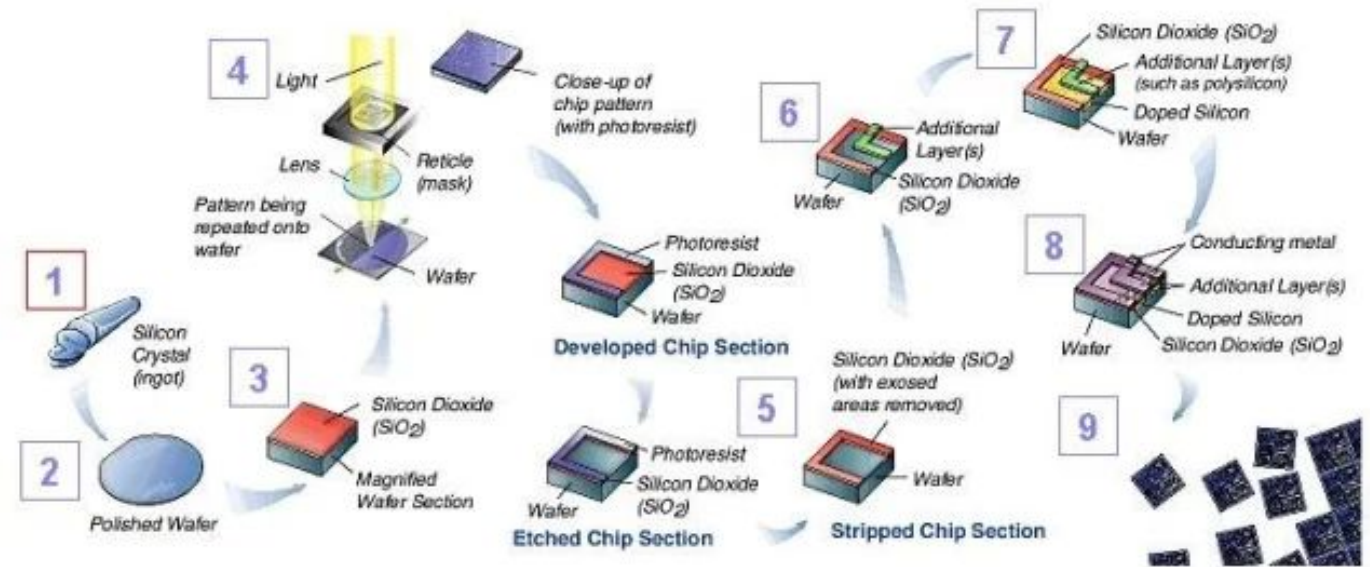
Fig. 6. Power Consumption of BERT in MIT Supercloud Dataset (Peak power consumption of approximately 48.70 kW, with an average consumption of 17.80 kW and a standard deviation of 12.39 kW. The job ran for an extended period of 4 days).

TAKEAWAYS FOR PRACTICE (ENERGY)

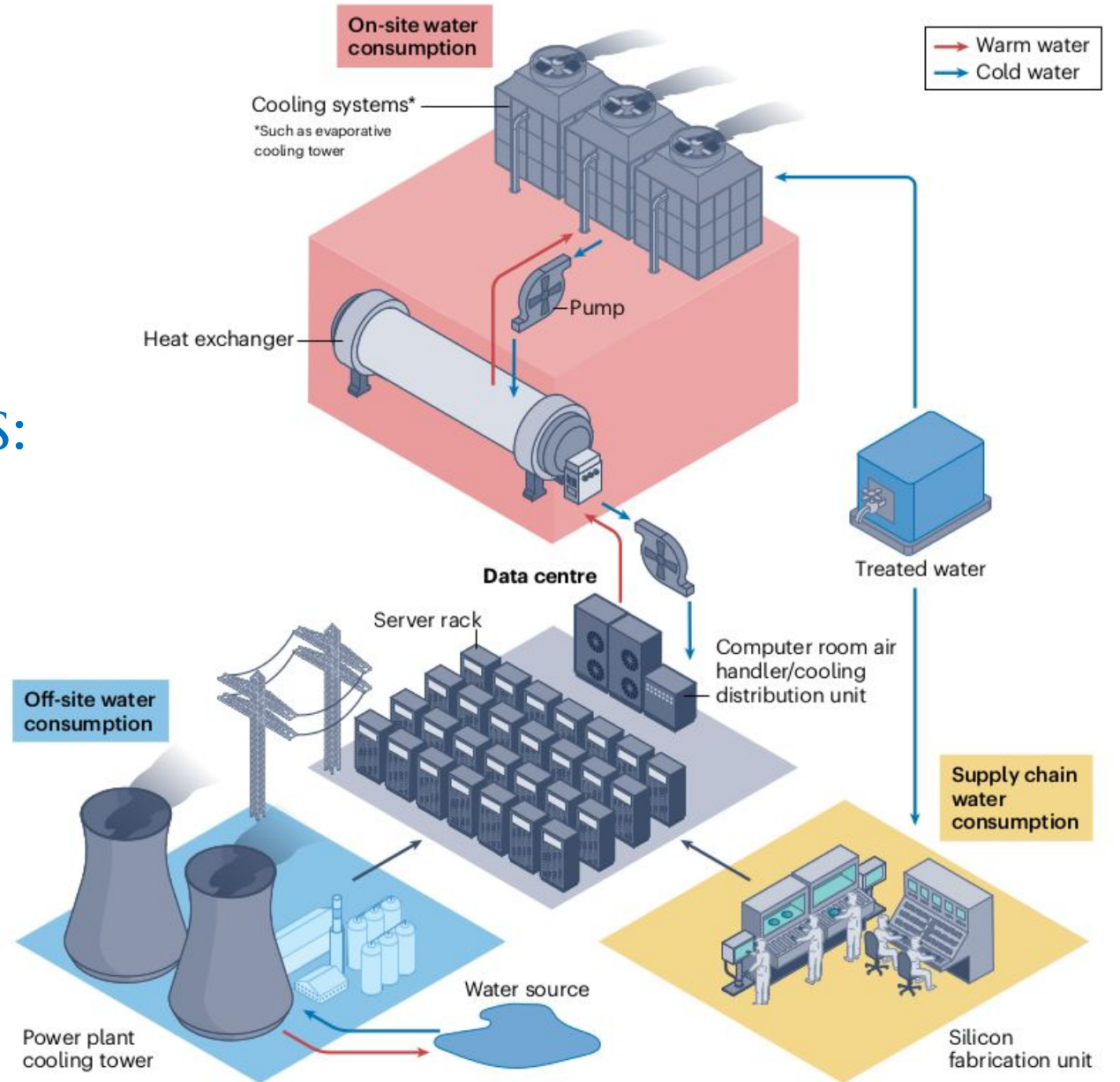
- Thinking about energy requirements of the models is important
- Not all models use a lot of energy! Big models vs small models 
- The consumption of the largest models doesn't discredit AI-for-climate as a whole
- Try to use a small / task-specific when possible: significant efficiency gains
- Think about which tasks are most impactful vs. potentially wasteful
- There are tools to compute the energy/GHG impacts of your project, e.g., [Code Carbon](#)

NON-GHG SUSTAINABILITY IMPACTS: RAW MATERIALS

Raw materials used to make computing chips come from 50 of our 90 naturally occurring elements here on Earth. For some, the stocks are getting rapidly depleted.



Non-GHG Sustainability Impacts: WATER



POLL FOUR

Is **direct** or **indirect** use of water for data centers larger?

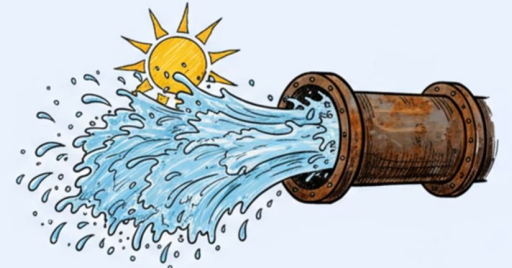
a) **direct**, b) **indirect**, c) **similar**

WATER: DIFFERENTIATED IMPACTS

- **Water withdrawal vs Water consumption vs Water discharge**
- About 50% consumption, direct currently less than 0.5% of US water consumption
- **Indirect use >>> Direct use** (up to 80%)
- Projected global scope 1 + 2 water withdrawal in 2027:
 - 4.2–6.6 billion cubic meters (more than 4-6x Denmark or ½ U.K.)
- **Local impact/stress:** Problem often more **peak** than annual totals;
- Bloomberg News reports that about two-thirds of U.S. data centers built since 2022 are in high water-stress areas.



The Bottle: **Annual Use**.
National totals seem modest and manageable on paper.



The Pipe: **Peak Demand**.
Massive withdrawals on hot days strain local infrastructure.

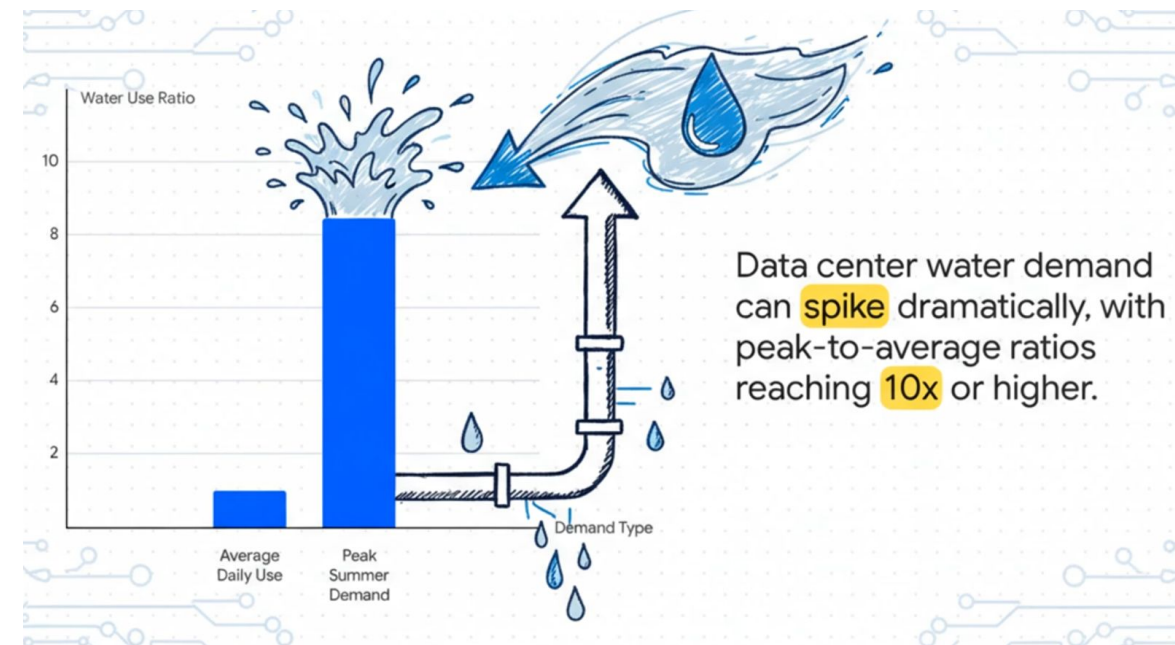
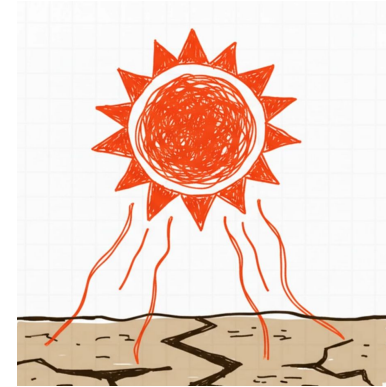
WATER: SOLUTIONS TO REDUCE NEEDS?

- “**Closed-loop**” liquid cooling (no evaporation) reduces water use to almost zero, but requires more energy (and can create more peaks)
- Increasingly possible to better **schedule** or **switch** between different cooling methods, but not necessarily for all parts of the data center
- **Indirect** consumption is more challenging. Depends on the choice of power technology (e.g., gas/nuclear >> solar) and on the local climate



WHEN IMPACTS ARE WORST... AT THE SAME TIME: **HEAT WAVES**

- **When** water consumption/withdrawal happens matters: worst during dry season / heat stress
- Same for energy consumption: during heat waves AC consumption stress the grid
- **Dilemma**: to cool down data centers, one can substitute water-intensive cooling and energy-intensive cooling, but both may be under pressure
- Water can shave peak loads but resource allocation is an issue; as many communities don't have the surplus water supply
→ **need for resilience planning**



SITUATED CASE STUDY: BRAZIL'S DATA CENTER EXPANSION

Massive ongoing wave of AI-driven infrastructure investments, exceeding R\$25 billion

89% of Brazil's electricity matrix is **renewable**, with almost half hydro-based → **ideal for data centers?**

Increasing droughts are putting pressure of water relevant both to hydropower generation and cooling, while access to **water is already an issue** for communities

Persistent **energy poverty**, over 1.3 million people still lack reliable electricity



Example: New data centers in Ceará and Rio Grande do Sul are threatening the access to water for local communities

LECTURE OUTLINE

Computing-related impacts

Application impacts

System level impacts

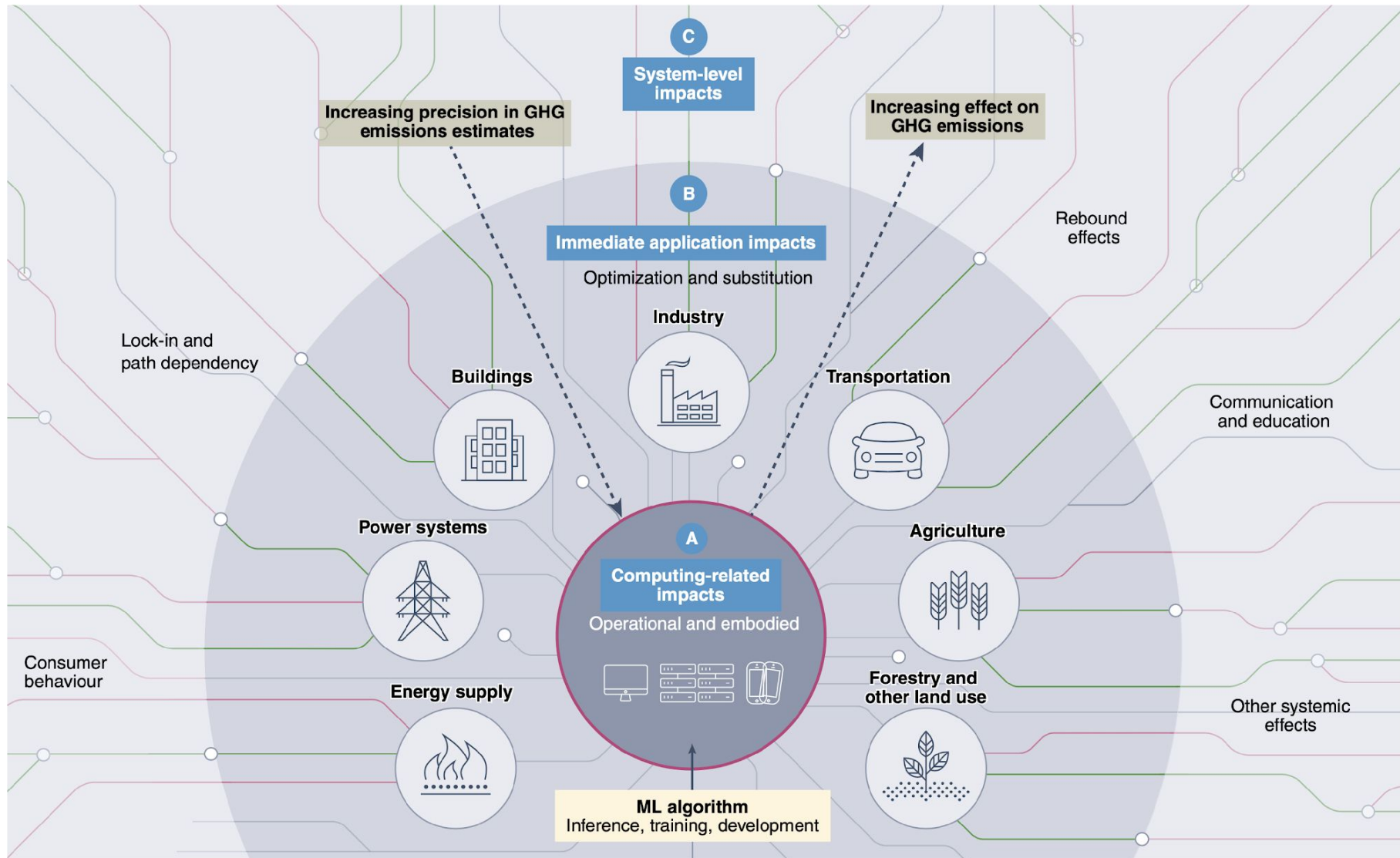
Overarching considerations

Regulating energy and resource usage of AI

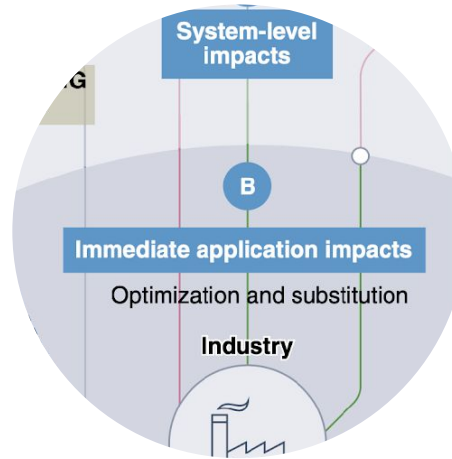
Positive impacts

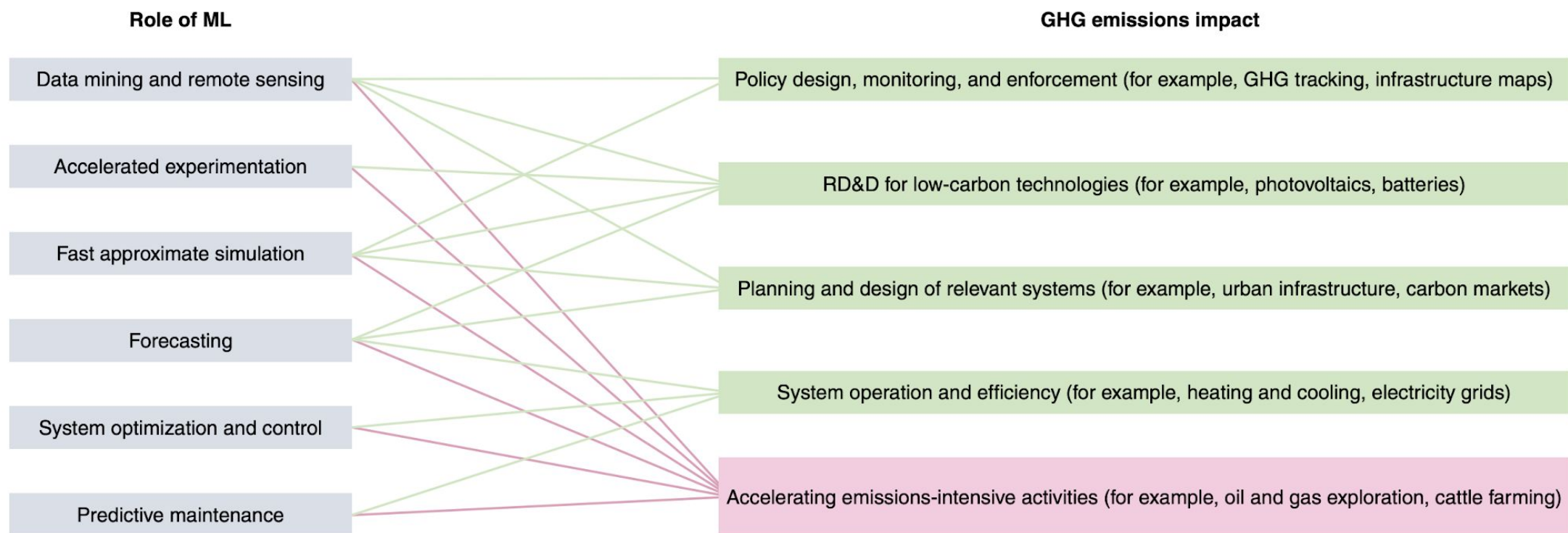
Negative
impacts

How AI Affects Greenhouse Gas Emissions

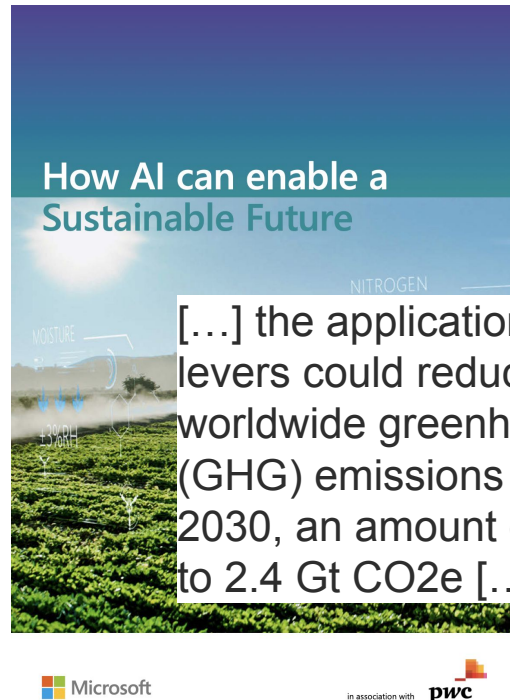


How AI Affects Greenhouse Gas Emissions

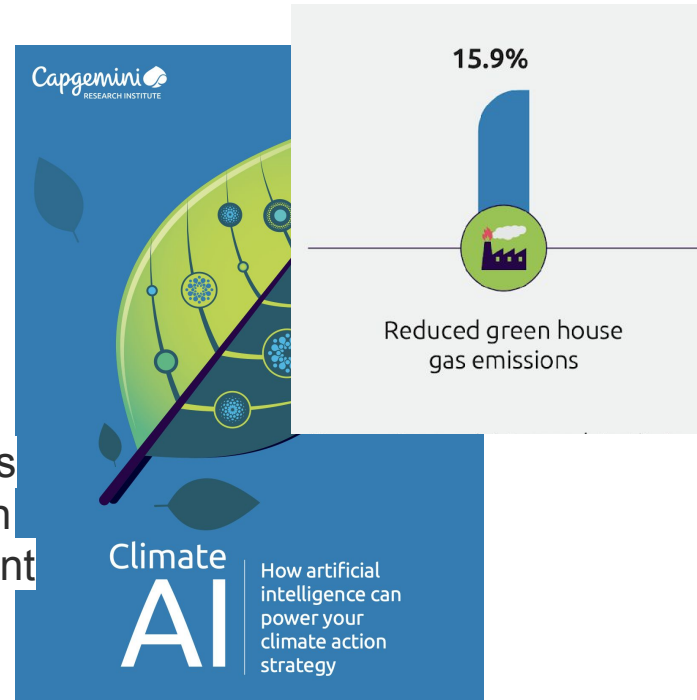




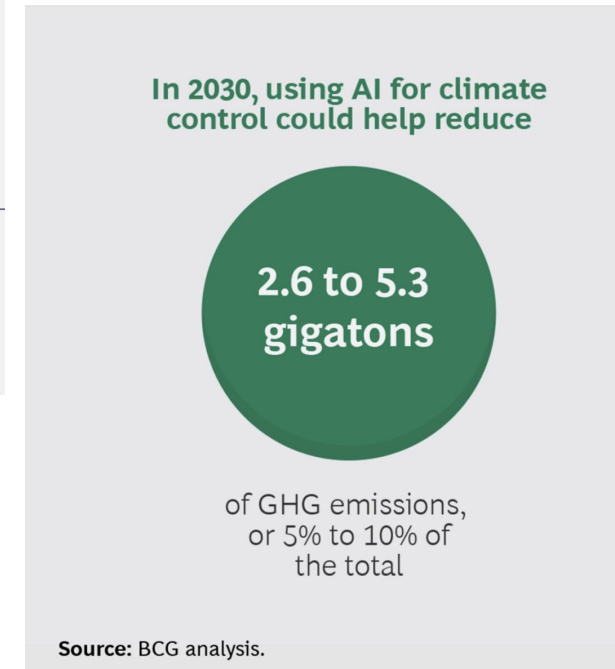
ESTIMATES OF THE GHG EMISSIONS SAVINGS POTENTIAL ARE UNRELIABLE



2019



2020



2021

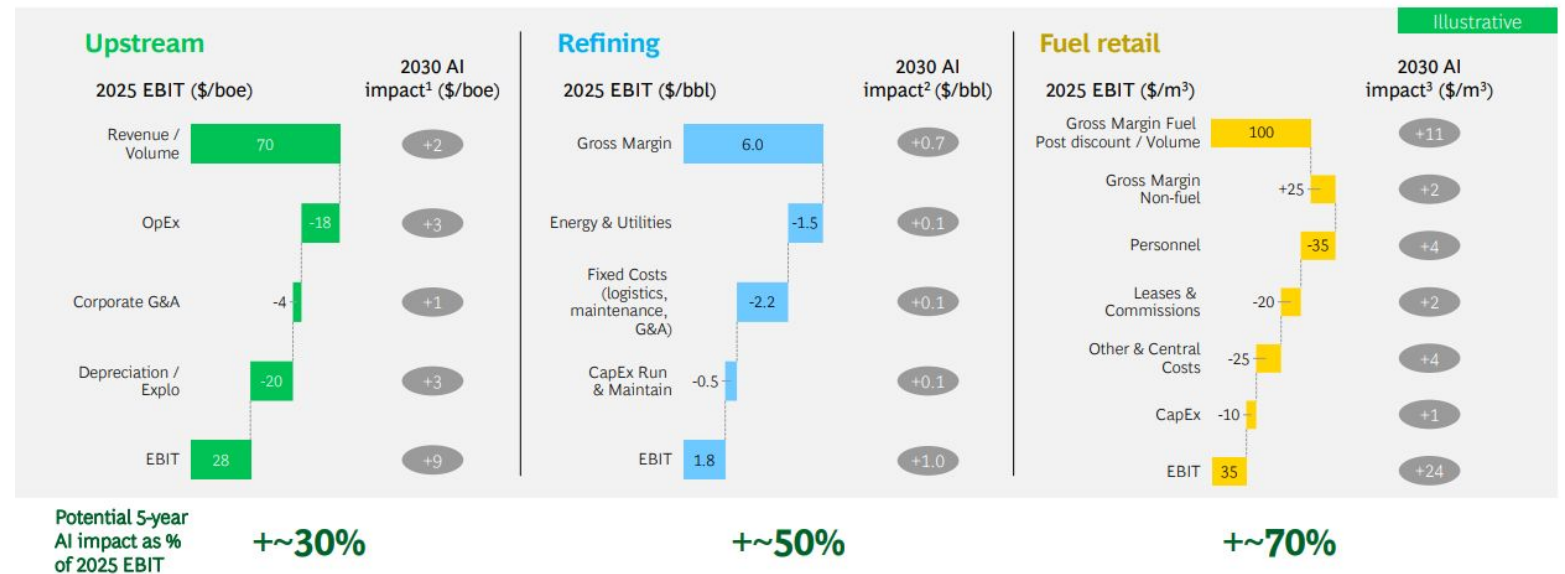
Estimates from consulting and tech firms often based on undisclosed modeling or extrapolation of stakeholder interviews → **vested interests likely overestimating the positive impacts**

AI APPLICATIONS INCREASING EMISSIONS

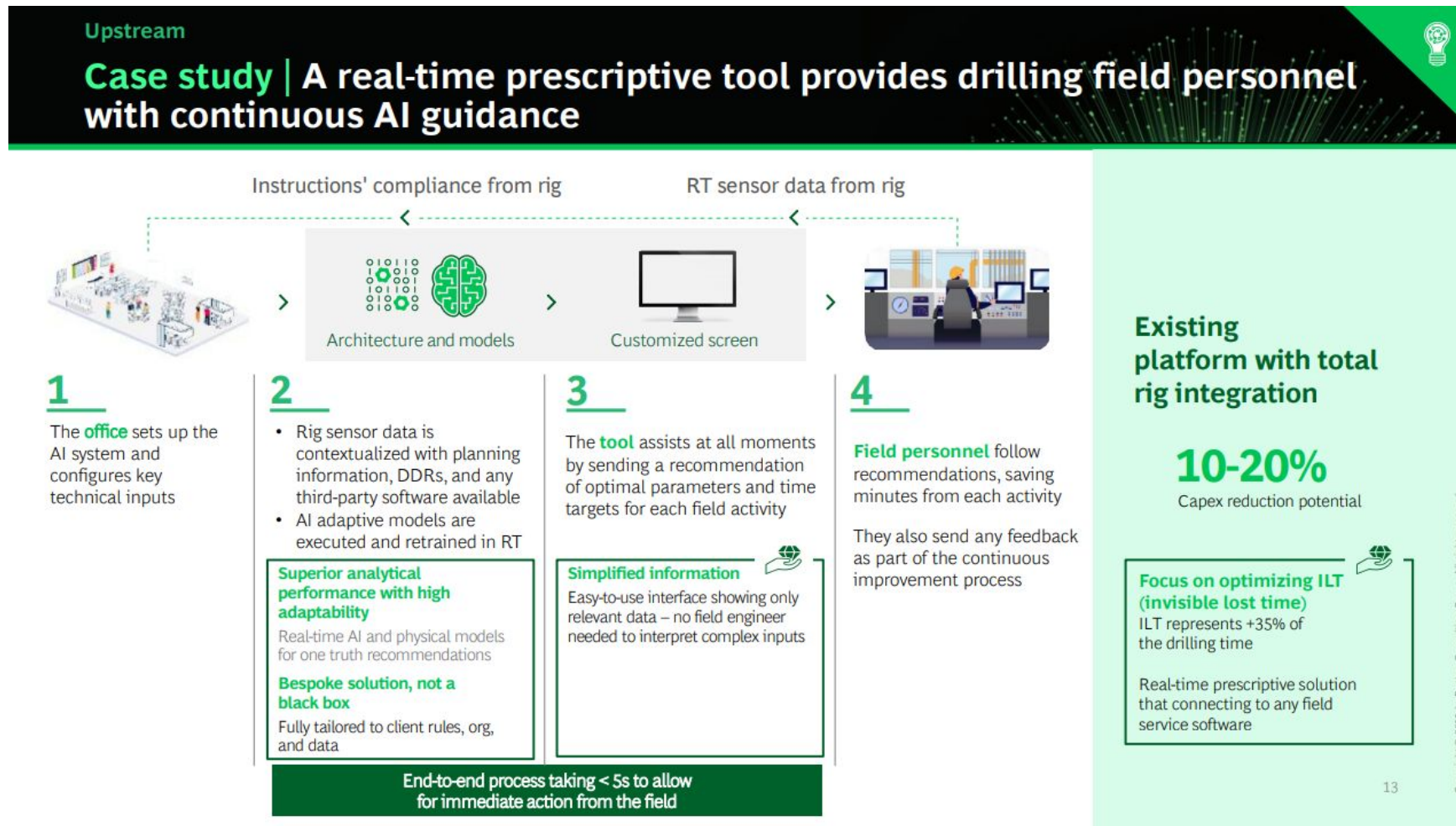


AI use to accelerate
emissions-intensive industries

Example: Oil and gas applications



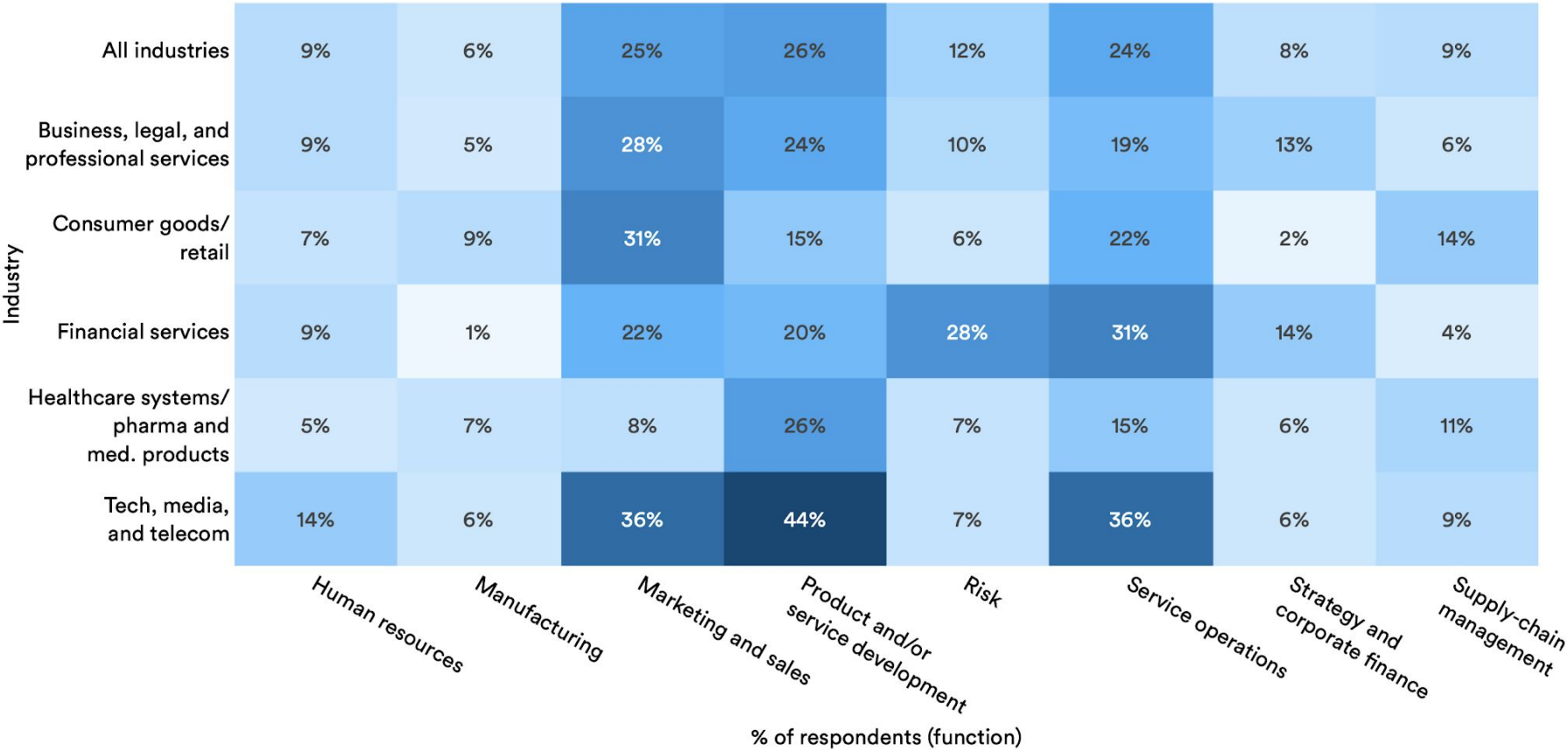
AI INCREASING IMPACTS EMISSIONS (AND BEYOND): OIL DRILLING



Most AI Use Across Society Is in Non-Climate Applications

AI adoption by industry and function, 2023

Source: McKinsey & Company Survey, 2023 | Chart: 2024 AI Index report



SO FAR, AI-FOR-CLIMATE IS RATHER TRADITIONAL AI

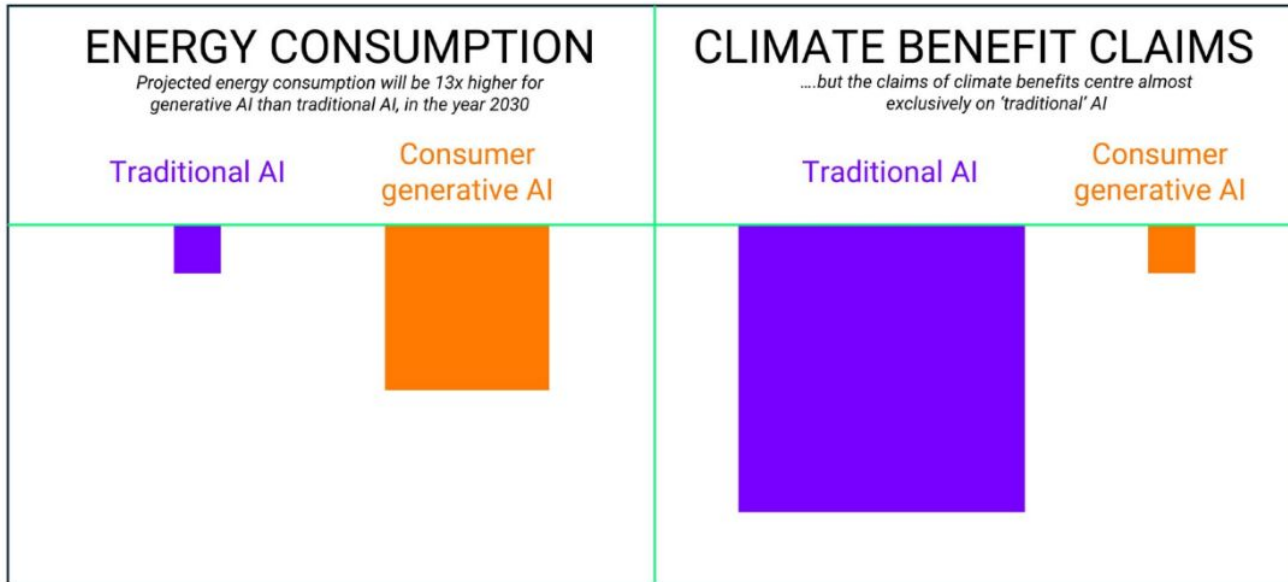


FIGURE 1 - CLIMATE BENEFIT versus DAMAGE

This graphic illustrates how most climate benefit claims relate to traditional AI, whereas most harm will relate to consumer generative AI. The first compares the projected electricity consumption in a central 2030 scenario, also represented by the area of each square. The second comparison shows the number of claims by AI type, represented by the area of each square.

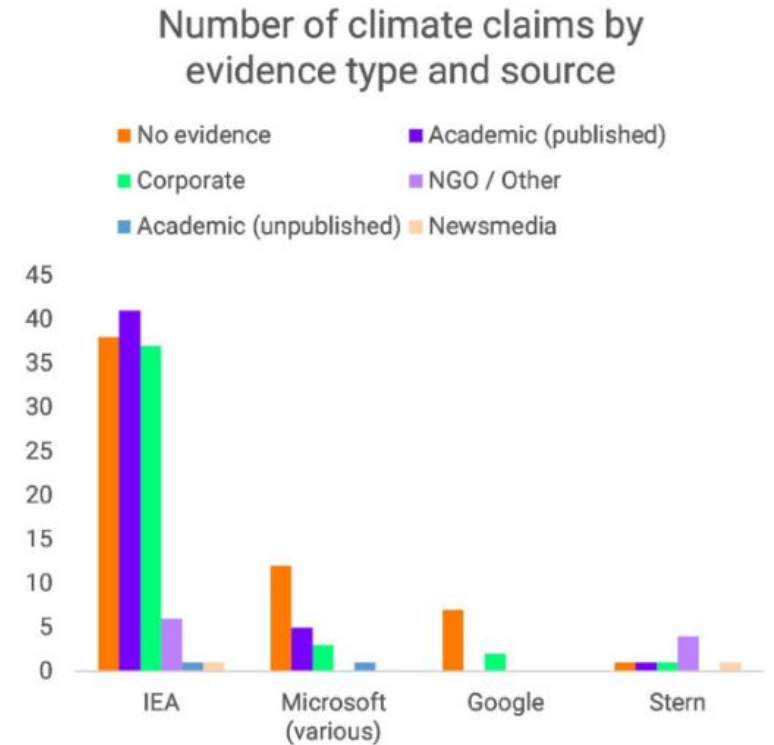


FIGURE 7 - TYPES OF EVIDENCE FOR CLIMATE CLAIMS

This graphic shows the distribution of evidence types by source. Corporate sources tended to rely on no evidence or corporate sites. The IEA's claims drew from a mix of academic, corporate and no evidence.

LECTURE OUTLINE

Computing-related impacts

Application impacts

System level impacts

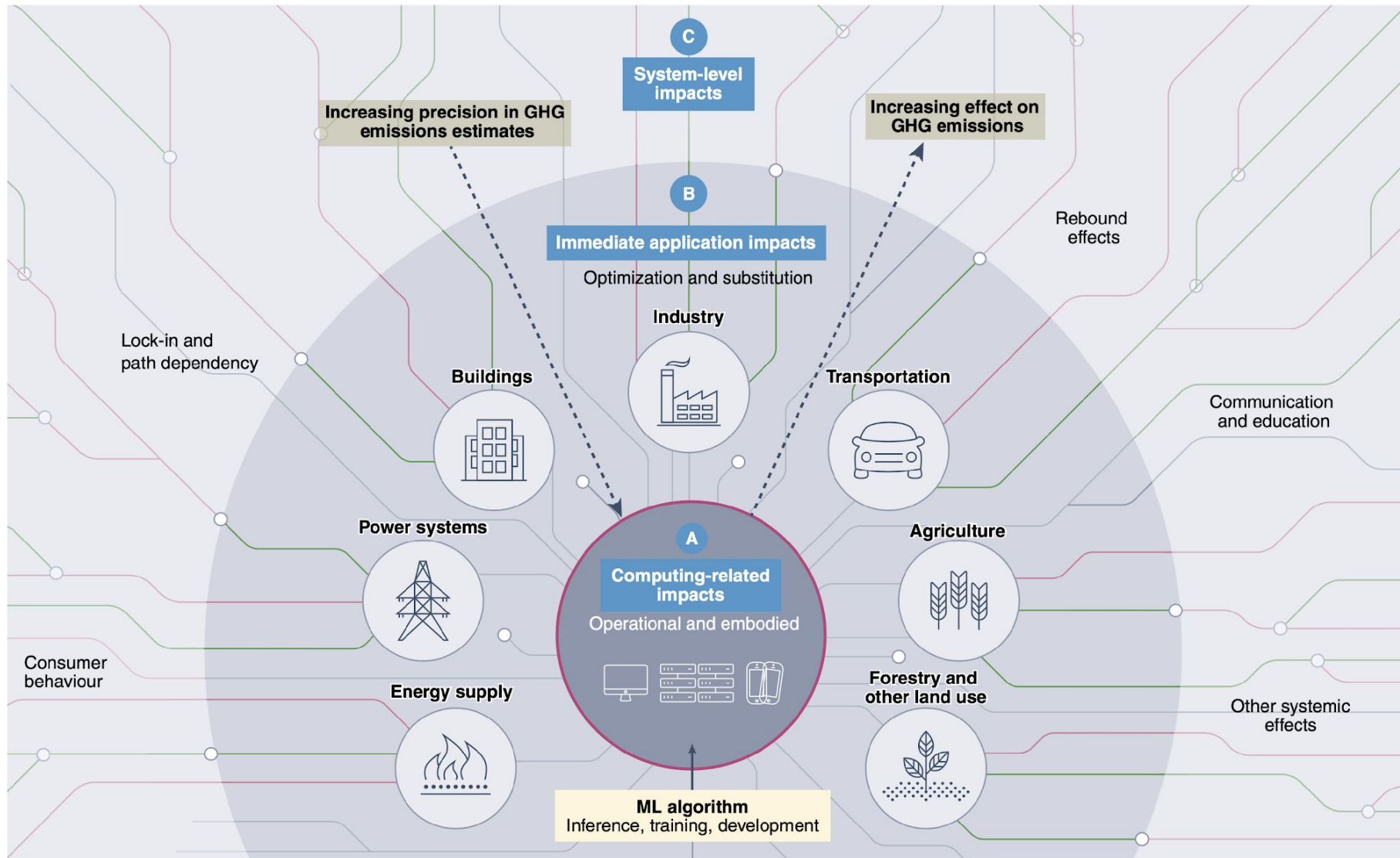
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Regulating energy and resource usage of AI

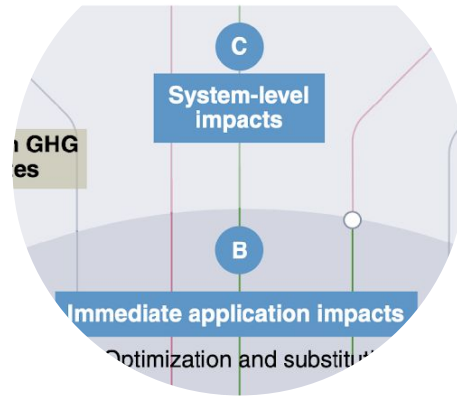
Rebound
effects

Societal power
shifts

How AI Affects Greenhouse Gas Emissions



How AI Affects Greenhouse Gas Emissions



SYSTEM-LEVEL IMPACTS OF AI APPLICATIONS

Potential **rebound** and **lock-in** effects

- Autonomous vehicles, ridesharing

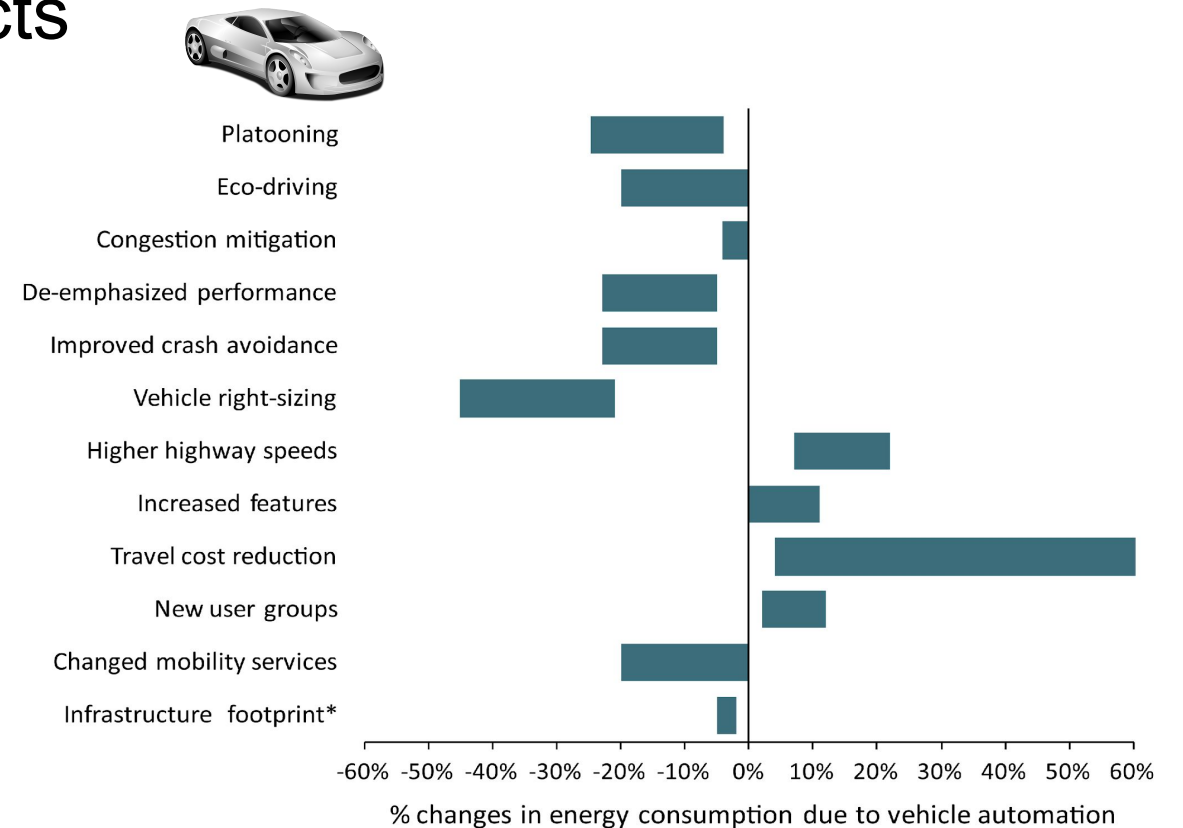


Image source: Wadud et al. 2016

SYSTEM-LEVEL IMPACTS OF AI APPLICATIONS

Potential **rebound** and **lock-in** effects

- Autonomous vehicles, ridesharing

Increasing **societal consumption**

- Personalized ads, on-demand delivery



Image credit: Megan_Rexazin_Conde / Pixabay.com

SYSTEM-LEVEL IMPACTS OF AI APPLICATIONS

Potential **rebound** and **lock-in** effects

- Autonomous vehicles, ridesharing

Increasing **societal consumption**

- Personalized ads, on-demand delivery

(Mis)information and **polarization**

- Content personalization/amplification



Image credit: Rose Wong / for NBC News

SYSTEM-LEVEL IMPACTS OF AI APPLICATIONS

Potential **rebound** and **lock-in** effects

- Autonomous vehicles, ridesharing

Increasing **societal consumption**

- Personalized ads, on-demand delivery

(Mis)information and **polarization**

- Content personalization/amplification

Inducing **societal power shifts**
due to access and agency

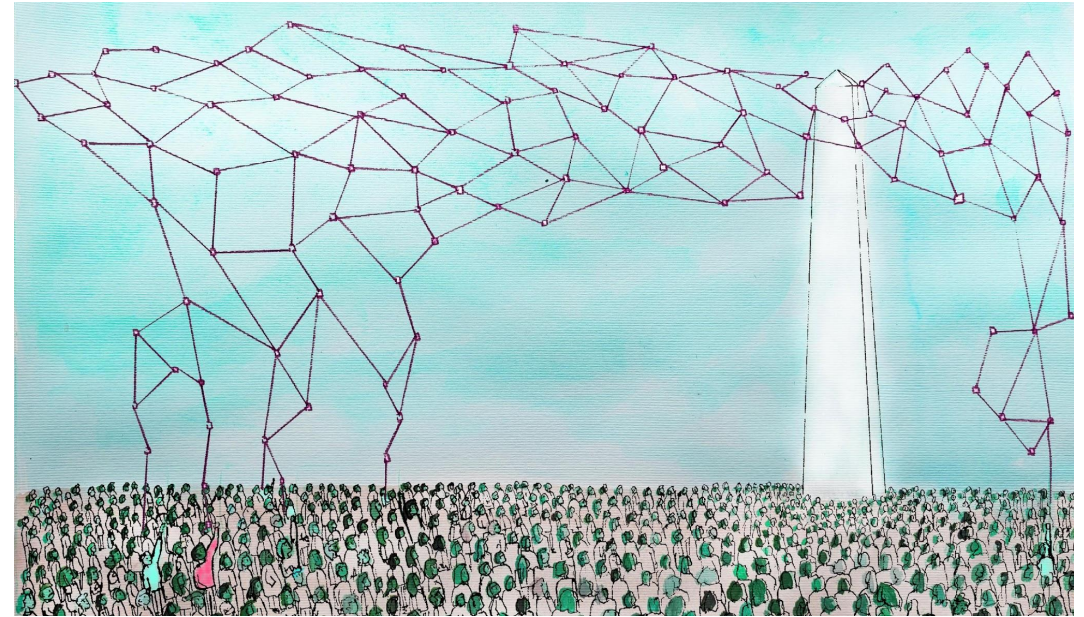


Image credit: Jamillah Knowles & We and AI / Better Images of AI / People and Ivory Tower AI / CC-BY 4.0

POLL FIVE

Share an example of a system-level impact of AI you believe has/will have a significant impact on climate, for one of the categories:

- a) rebound effect, b) increased consumption,
- c) mis-information and polarization, d) societal power shifts

LECTURE OUTLINE

Computing-related impacts

Application impacts

System level impacts

Overarching considerations

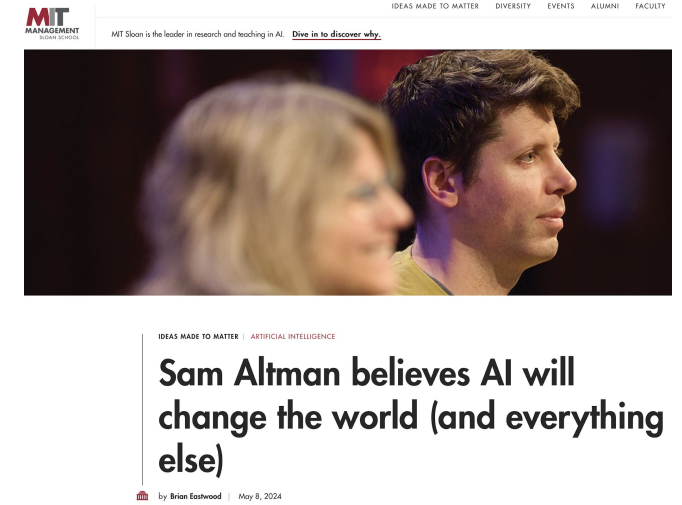
Regulating energy and resource usage of AI

Narratives

Estimates

THE “GENAI FALLACY” : USING AI-FOR-CLIMATE AS JUSTIFICATION FOR LARGE MODELS

- To justify the energy consumption in datacenters for large models such as LLMs, AI-for-climate applications are a **welcome narrative** for tech companies
- But, most AI applied in climate-relevant sectors is not large models
- We can have AI-for-climate progress **without** large computing-related energy costs



“If we have to spend even 1% of the world’s electricity training powerful AI, and that AI helps us figure out how to get to non-carbon-based energy or to do carbon capture better, that would be a massive win”

Misleading statement from Sam Altman

“CHRONIC POTENTIAL -ITIS”

- Concept from Vlad Coroamă: lot of discussion about potential...
- But there are significant barriers to AI deployment in climate-related domains
- Actors are presenting AI's savings potentials by constructing **best case adoption scenarios**
- Visualizations and summaries communicate **overconfidence** in reaching hypothetical estimates

Key
messages
from the 2025
Shell Energy
Security
Scenarios



1. In a world where artificial intelligence (AI) brings societal change and catalyses economic growth, demand for energy increases and the transition to low-carbon energy accelerates. ✓

HOLISTIC ESTIMATES?

Source	Type	Computing-related	Application-related (reductions)	Application-related (increases)	System-level
Stern et al. 2025	Journal article	Total data center increase	Power, meat and dairy, and light road vehicles	No	No
IEA 2025	Technical report	Total data center	Some energy applications	No	Only rebound
PWC 2025	Consulting report	Total data center increase	Economy-wide	No	No
Schneider Electric 2024	Technical report	AI	No	No	No
de Vries 2023	Journal article	AI	No	No	No

No application-related increases & no system-level effects assessed

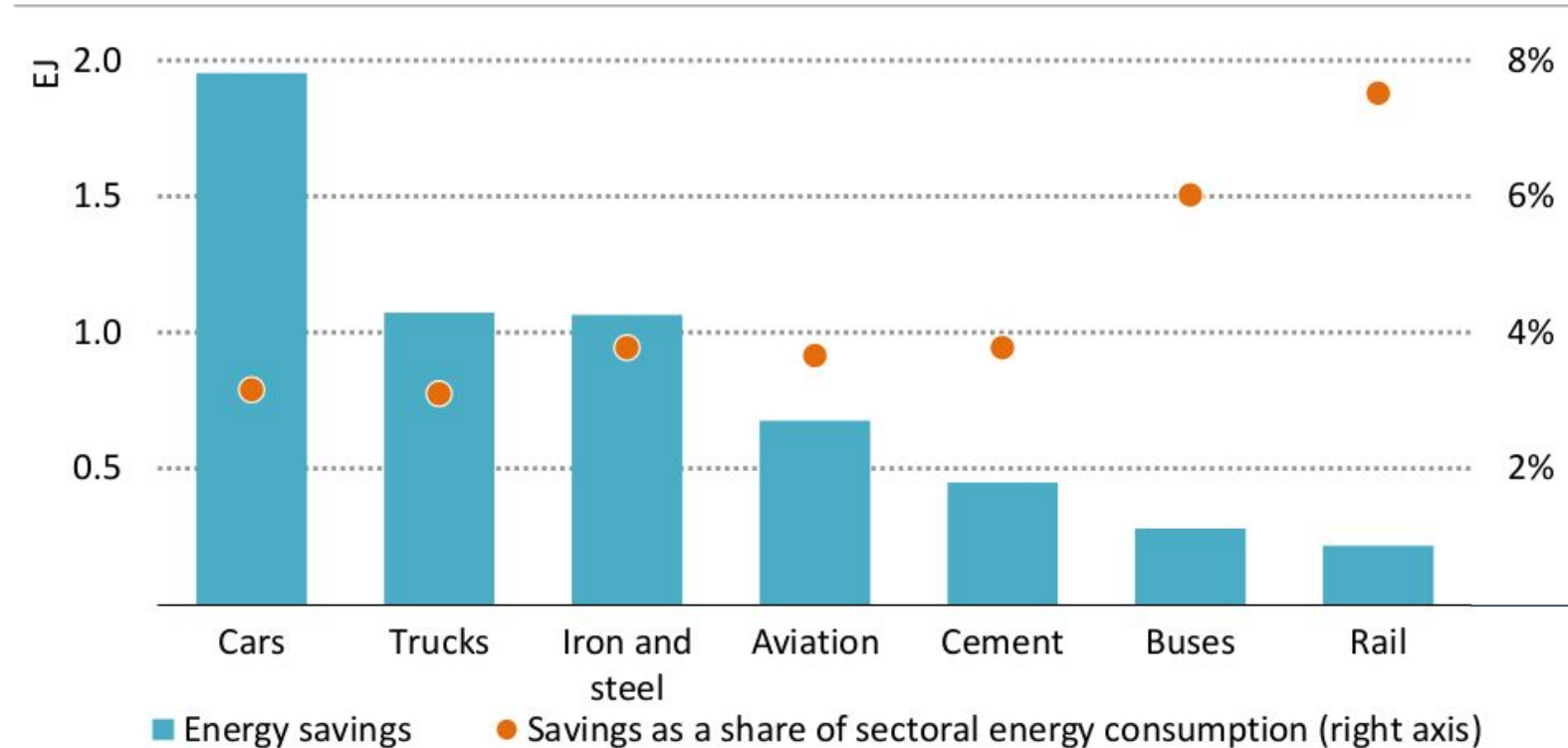
DANGER OF FALLACIES IN (LACK OF) COMPARISONS BETWEEN POSITIVE & NEGATIVE IMPACTS OF AI

"is AI's energy use
worth the climate
benefits?"
is an **ill-posed** question

potential energy
savings in certain
sectors

≠ net climate benefit

≠ a major global
accelerant in achieving
global climate goals.



IEA. CC BY 4.0.

Widespread adoption of known AI applications has the potential to save 13.5 EJ across all end-use sectors in 2035, greater than Indonesia's annual energy consumption today

Note: This energy savings potential is estimated using existing AI applications only and does not factor in solutions that might emerge in the future. A detailed discussion of this approach is available in IEA (2025a).

Widespread
adoption
of AI across
industries

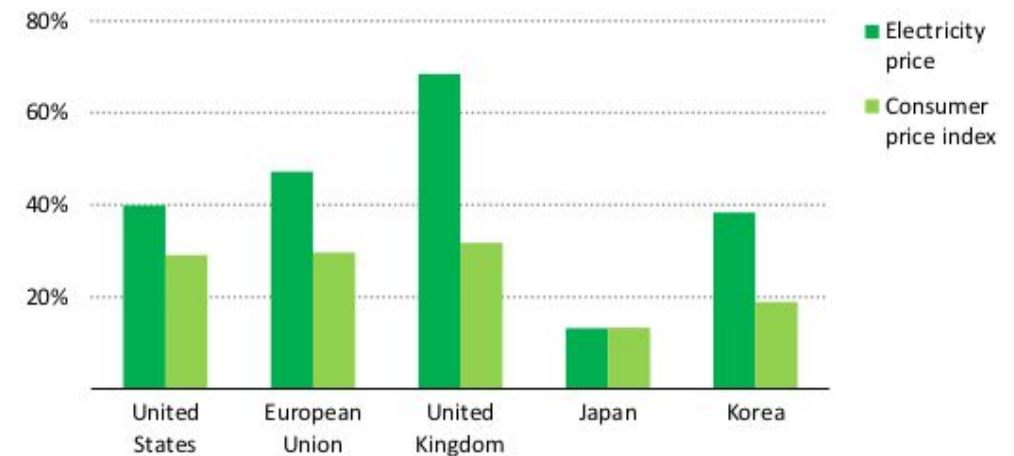
GREENING THE GRID IS NOT SUFFICIENT

- Unbridled AI growth is not something that can be handled by the Planet
- has social impacts already

The path to net zero emissions is narrow: staying on it requires immediate and massive deployment of all available clean and efficient energy technologies. In the net zero emissions pathway presented in this report, the world economy in 2030 is some 40% larger than today but uses 7% less energy. A major worldwide push to increase energy efficiency is an essential part of these efforts, resulting in the annual rate of energy intensity improvements averaging 4% to 2030 – about three-times the average rate achieved over the last two decades.

Source: IEA, “Net Zero by 2050: A Roadmap for the Global Energy Sector” (2021)

Figure 8.1 ▶ Change in nominal residential electricity price and consumer price index by country or region, January 2019–December 2025



Residential electricity prices have risen faster than headline inflation in several markets in recent years

IEA. CC BY 4.0.

Source: IEA, “Energy and AI” 2026

LECTURE OUTLINE

Computing-related impacts

Application impacts

System level impacts

Overarching considerations

Regulating energy and resource usage of AI

Fostering
AI-for-climate

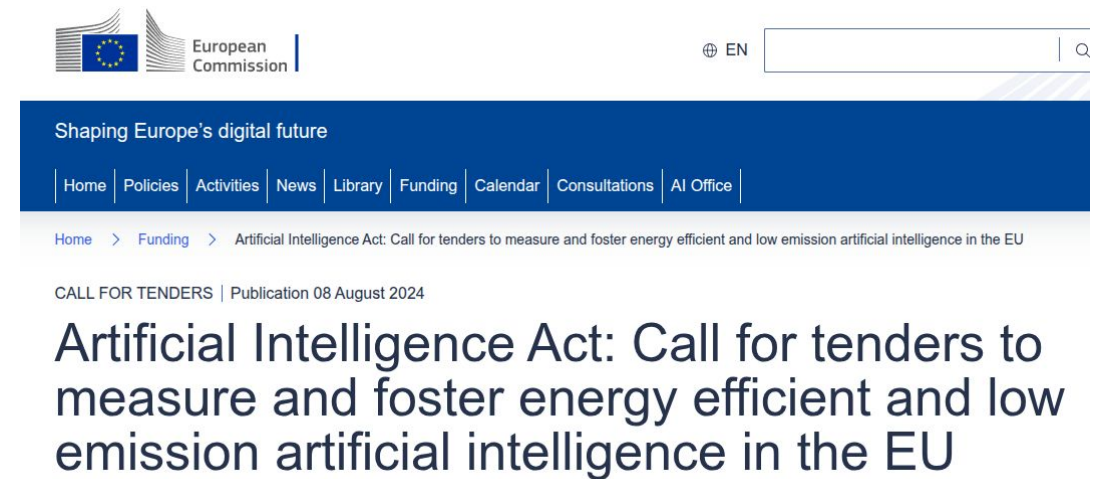
Requiring
transparency

CALLBACK: AI & CLIMATE POLICY AND REGULATION LECTURES

TODO: add summary slides if i can get access to the
other decks

EXISTING INITIATIVES TO FOSTER APPLICATIONS

- ♦ Dedicated governmental research and development funding (EU, Germany, Austria, Sweden, US, etc.)
- ♦ Large research and monitoring programs (Copernicus, Nasa Harvest, etc.)
- ♦ Some AI strategies and EU AI Act (2024) emphasize climate change as application area



INCENTIVIZING FRUGAL APPLICATIONS



General framework for frugal AI

Measuring and reducing the environmental impact of AI

IA et environnement : 12 territoires sélectionnés pour concevoir l'IA frugale au service de la transition écologique

Publié le 13 septembre 2024 , par **Olivier Devillers**, pour Localtis

Environnement, Numérique

La Banque des Territoires a accueilli le 10 septembre 2024 les 8 nouveaux lauréats de l'appel à projets démonstrateur d'IA frugale pour la transition écologique des territoires, rejoignant les 4 territoires déjà labellisés. L'occasion de découvrir, avec des exemples variés et concrets, la notion de frugalité appliquée à l'intelligence artificielle.



EXISTING INITIATIVES REQUIRING TRANSPARENCY

- ♦ The EU AI Act (2024)
 - Reporting on the compute resources and energy use of high-risk systems and large models
- ♦ Proposed US Artificial Intelligence Environmental Impacts Act (2024)
 - Require environmental impacts assessments (by the EPA and NIST)
 - Application- and computing-related impacts



Official Journal
of the European Union

EN
L series

2024/1689

12.7.2024

REGULATION (EU) 2024/1689 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

of 13 June 2024

laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act)

(Text with EEA relevance)

Annex IV: Technical Documentation

[for high-risk AI systems]

(c) [...] the computational resources used to **develop, train, test and validate** the AI system;

Annex XI: Section 1 – Information to be provided by all providers of general-purpose AI models

(d) the computational resources used to **train** the model (e.g. number of floating point operations), training time, and other relevant details related to the training;

(e) **known or estimated energy consumption of the model**. With regard to point (e), where the energy consumption of the model is unknown, the energy consumption may be based on information about computational resources used.

POLL SIX

The EU AI Act requires public disclosure for the energy use for general-purpose AI required for:

a) training, b) inference, c) both, d) none

OMISSIONS MATTER

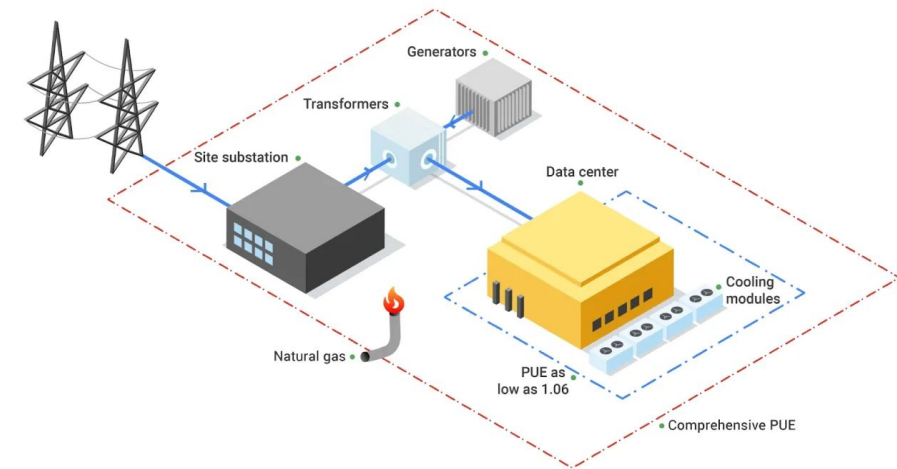


Example in the **EU AI act**:

- For **general-purpose AI**: significant gap as it only covers the energy used during the model's development phase, but **leaves out the inference phase**
- Where the Act does mandate the disclosure of energy consumption, this **information is restricted to authorities and is not accessible to downstream providers** → significantly reduces transparency and accountability, thereby weakening the potential for public oversight and market responses.
- AI Act fails to address the greenhouse gas (GHG) emissions generated **by** AI applications
- Water consumption not addressed
- Environment can in theory be regarded as a criterion for high-risk systems but it is not made explicit.

REGULATING DATA CENTERS

- Energy Efficiency Directive EU/2023/1791 ("EED") and the Commission Delegated Regulation EU/2024/1364
 - applies to all data centers in the EU with a power demand of at least 500kW, which includes small-sized data centers.
 - Data center operators are required to collect, make publicly available and report to a EU database information. The reporting is mandated on an annual basis.
 - The required data includes energy consumption, power utilization, temperature set points, waste heat utilisation, water usage and use of renewable energy
- German 2023 Energy Efficiency Act
 - requires data center operators to inform their customers on an annual basis on the energy consumption directly



Power Usage Effectiveness
(Image: Google)

NEED FOR INTEGRATED OVERSIGHT FOR DATACENTER SITTING

- Transform data-infrastructure governance from a purely energy-water challenge into a lever for equitable territorial planning and people's centred policy vision.
- Land use interaction: Siting has various implications given interaction with local zoning, property valuation, and ecosystem restoration priorities
- Ongoing discussion on compensation mechanisms, and community benefit frameworks needed
- → **Need for integrated oversight across energy, environment, and digital-economy policy vision.**



SUMMARY

 Sustainability impacts of AI span the whole AI lifecycle and include various impact categories

→ Look at computing, applications and system-level impacts

 Impacts of large models need to be better addressed via evidence-based action

 Yes to frugal AI-for-climate applications!

 Estimates remain uncertain and outlooks/narratives should be cautious

 Regulating AI, beyond market incentives, can lead to further transparency, which is key to public debate

MORE RESOURCES

Inventory of Reports on Sustainable AI

Explore a curated selection of reports on sustainable AI, published by our members. This page showcases key publications that shed light on the latest advances, challenges, and best practices in the design, deployment, and environmental impact of artificial intelligence.

The opinions expressed in these reports are those of the issuing organization and do not necessarily reflect the views of the Coalition's members.

Key Reports & Publications on Green AI & AI for Green

Energy Consumption

Frugal AI

Environmental Impact & Metrics

Transparency

Water Consumption

Climate Action

Biodiversity

Ocean

<https://www.sustainableaicoalition.org/inventory-of-key-reports-on-sustainable-ai/>

DATA

Data Center Policy Database

Built For Policymakers, Researchers, Journalists, And Communities.

[View Database](#)[Read Our Blog](#)[About Us](#)

Independent Statistics and Analysis
U.S. Energy Information Administration

[Sources & Uses](#) | [Topics](#) | [Geography](#) | [Tools](#) | [Education](#) | [News](#)

[Electricity Data](#)

Electricity Data Browser



Explore the new **Beta version** with expanded plant level data for water cooling and emissions.

Change data set

Net generation

OR

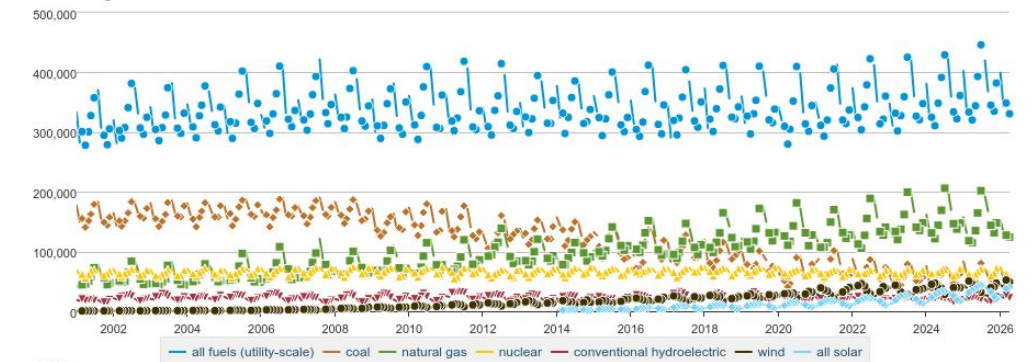
View a pre-generated report

Choose a report

Net generation, United States, all sectors, monthly

DOWNLOAD

thousand megawatthours



Data source: U.S. Energy Information Administration

CHART INDEXING OPTIONS: **None** | [Index to start as percent](#) | [Index to start as value](#)

Q&A

QUIZ ONE

Is the energy use for training or inference most important for the main commercial large language models for

a) Training, b) Inference?

QUIZ Two

Are larger models always performing better than smaller models in terms of results for climate-relevant tasks?

a) True, b) False

QUIZ THREE

Do we have robust scientific estimates about the *future* net impact of AI on climate?

a) True, b) False

QUIZ FOUR

Qualitatively,
is the *current* net impact of AI on climate:
a) Negative, b) Positive?

QUIZ FIVE

Is using more renewable energy the solution to the increasing electricity demand of AI?

a) True, b) False

PRE-READINGS

- Aligning artificial intelligence with climate change mitigation, Kaack et al 2022
<https://www.nature.com/articles/s41558-022-01377-7>
- The AI Climate Hoax: Behind the Curtain of How Big Tech Greenwashes Impacts, Ketan Joshi, 2026
https://drive.google.com/file/d/12l1W4W25b-_ff6yFNJABkfal9_9oevxe/view
- The Real Story on AI's Water Use—and How to Tackle It, Shaolei Ren & Amy Luers, 2025 <https://spectrum.ieee.org/ai-water-usage>